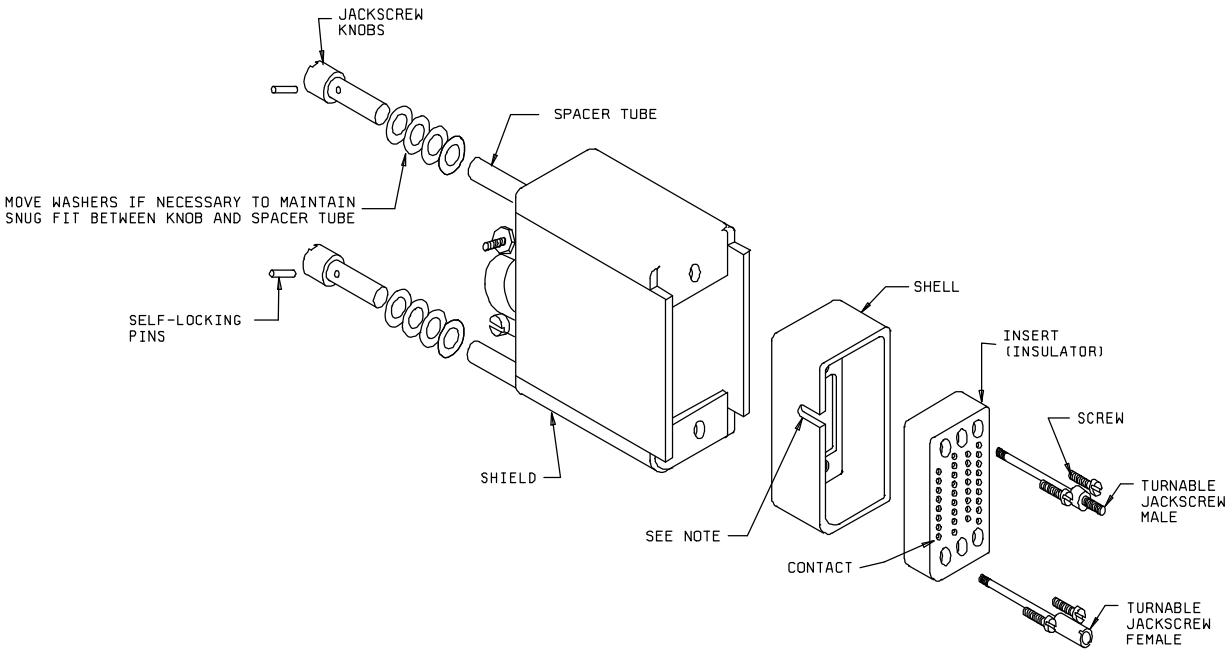
INCH-POUND
MIL-DTL-28748/2D
11 May 1999
SUPERSEDING
MIL-C-28748/2C
30 November 1981

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, SOLDER TYPE SOCKET CONTACTS, SIZE 16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue MIL-DTL-28748 listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:



Connector

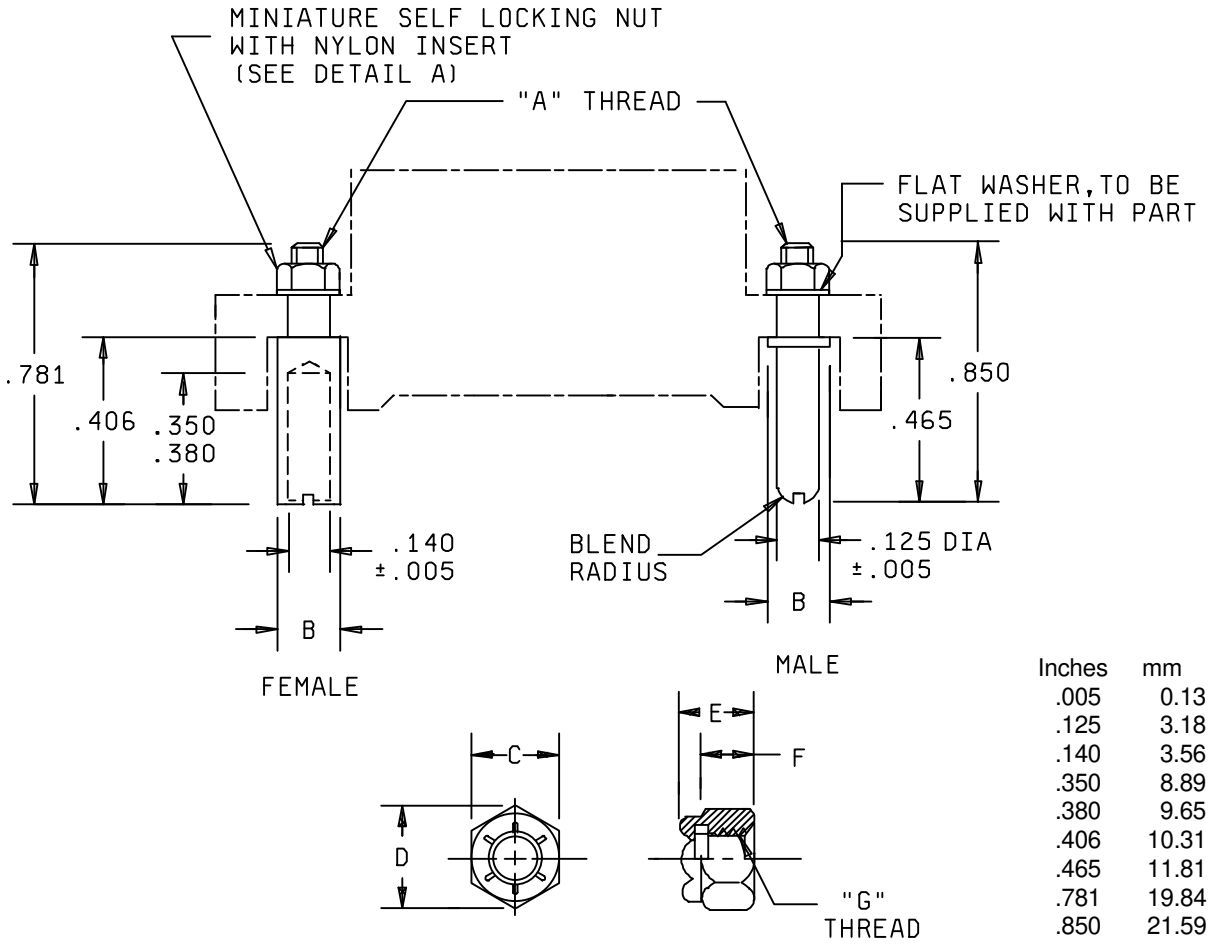
NOTE: The polarization feature (groove on the receptacle shell, pin on the plug shell) shall be adjacent to the side closest to the number 1 or A contact position.

Configuration A

FIGURE 1. Connector with shield, plug shell, jackscrews, and guidepins.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Inches	mm
.005	0.13
.125	3.18
.140	3.56
.350	8.89
.380	9.65
.406	10.31
.465	11.81
.781	19.84
.850	21.59

Detail A
Miniature self locking nut
(nylon insert)

Guidepin designator G

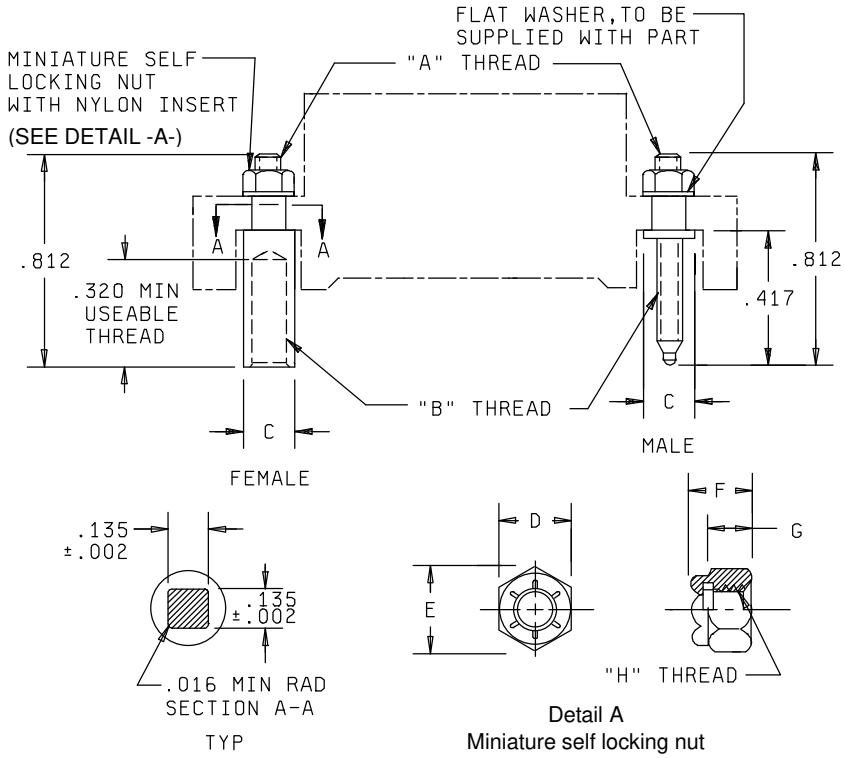
Style	Connector size (number of contacts)	A Thread class 2	B dia max	C min	C max	D ref	E min	E max	G Thread class 3
Male	9, 14, 20, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	.190 (4.83)	.181 (4.06)	.190 (4.83)	.205 (5.21)	.095 (2.41)	.125 (3.18)	4-40 UNC-3B .112-40 UNC
Female	9, 14, 20, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	.190 (4.83)	.181 (4.06)	.190 (4.83)	.205 (5.21)	.095 (2.41)	.125 (3.18)	4-40 UNC-3B .112-40 UNC
Male	104	6-32 UNC-2A .138-32 UNC	.190 (4.83)	.240 (6.10)	.252 (6.04)	.275 (6.99)	.133 (3.38)	.153 (3.89)	6-32 UNC-3B .138-32 UNC
Female	104	6-32 UNC-2A .138-32 UNC	.190 (4.83)	.240 (6.10)	.252 (6.04)	.275 (6.99)	.133 (3.38)	.153 (3.89)	6-32 UNC-3B .138-32 UNC

Connector with guidepins

Configuration B

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Detail A
Miniature self locking nut
(nylon insert)

Inches	mm
.002	0.05
.016	0.41
.135	3.43
.320	8.13
.417	10.57
.812	20.62

Jackscrew designator F

Style	Connector size (number of contacts)	A Thread class 2	B Thread class 2	C dia max
Male	9, 14, 20, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	6-32 UNC-2A .138-32 UNC	.190 (4.83)
Female	9, 14, 20, 34, 42, 50, 66, 75	4-40 UNC-2A .112-40 UNC	6-32 UNC-2B .138-32 UNC	.190 (4.83)
Male	104	6-32 UNC-2A .138-32 UNC	6-32 UNC-2A .138-32 UNC	.225 (5.72)
Female	104	6-32 UNC-2A .138-32 UNC	6-32 UNC-2B .138-32 UNC	.225 (5.72)

Self locking nut

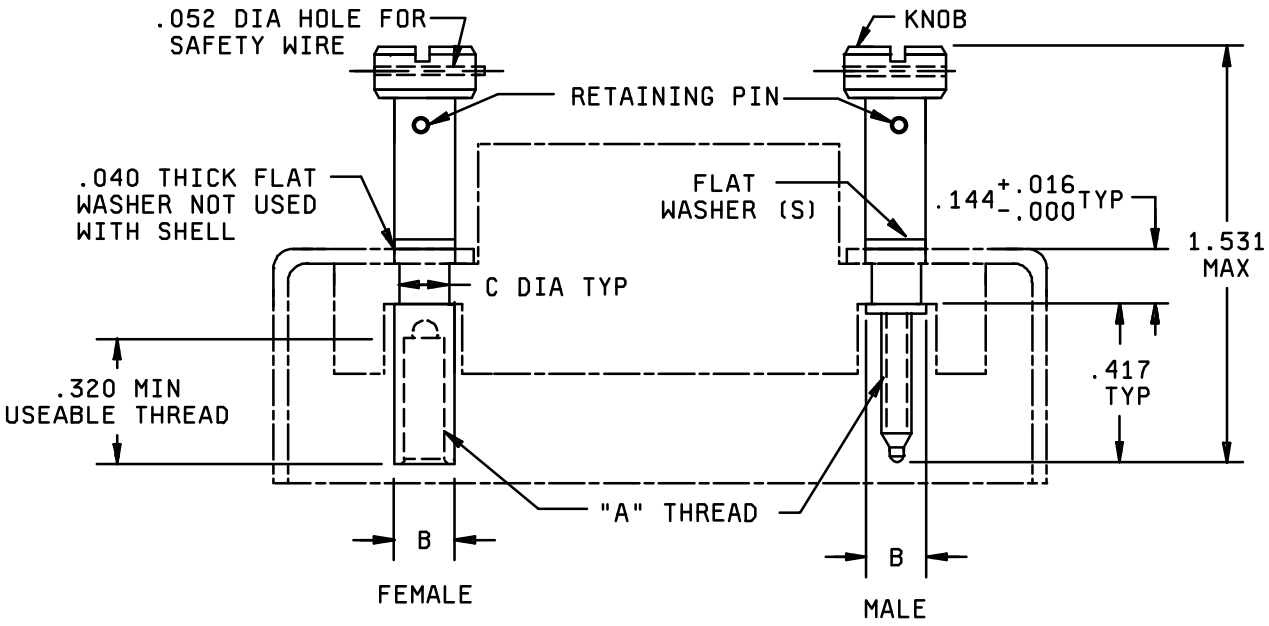
Connector size (number of contacts)	D		E ref	F		G ref	H Thread class 3
	min	max		min	max		
9, 14, 20, 34, 42, 50, 66, 75	.181 (4.06)	.190 (4.83)	.205 (5.21)	.095 (2.41)	.125 (3.18)	.070 (1.78)	4-40 UNC-3B .112-40 UNC
104	.240 (6.10)	.252 (6.04)	.275 (6.99)	.133 (3.38)	.153 (3.89)	.081 (2.06)	6-32 UNC-3B .138-32 UNC

Connector with fixed jackscrews

Configuration C

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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Inches	mm
.016	0.41
.040	1.02
.052	1.32
.144	3.60
.320	8.13
.417	10.59
1.531	58.89

Jackscrew designator S

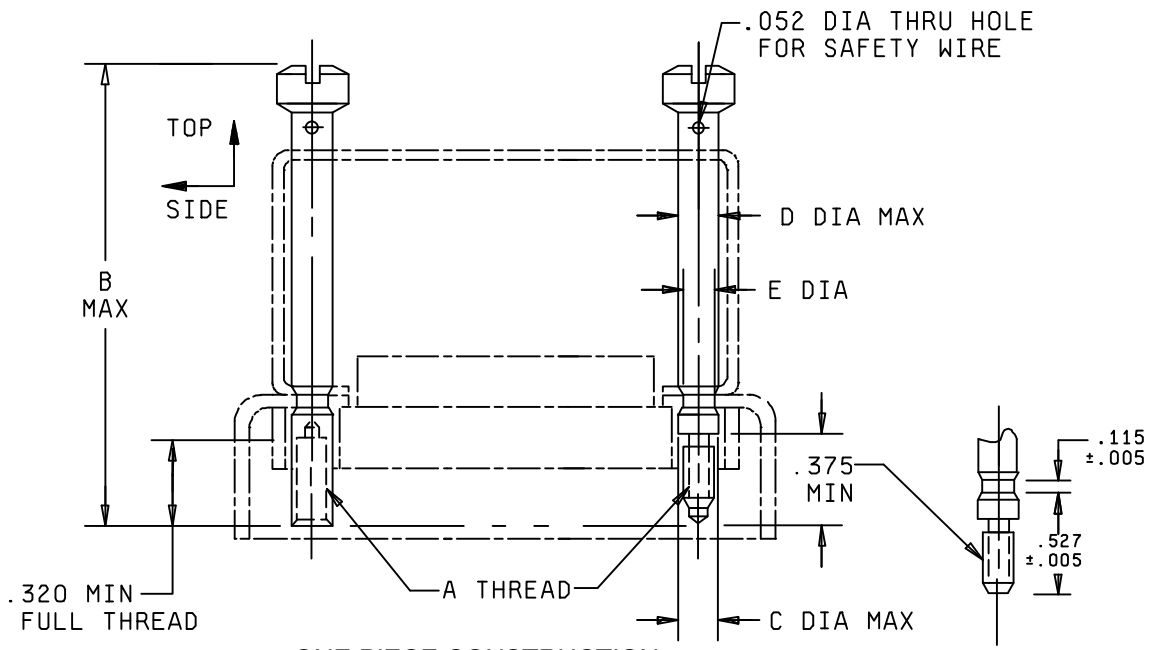
Style	Connector size (number of contacts)	"A" Thread class 2	"B" dia max	C DIA ±.003 (0.08 mm)
Male	7, 9, 14, 20, 26, 34, 42, 50, 66, 75	6-32 UNC-2A .138-32 UNC	.190 (4.83)	.115 (2.92)
Female	7, 9, 14, 20, 26, 34, 42, 50, 66, 75	6-32 UNC-2B .138-32 UNC	.190 (4.83)	.115 (2.92)
Male	104	8-32 UNC-2A .164-32 UNC	.225 (5.72)	.137 (3.48)
Female	104	8-32 UNC-2B .164-32 UNC	.225 (5.72)	.137 (3.48)

Connector with short turnable jackscrews

Configuration D

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

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ONE PIECE CONSTRUCTION

Inches	mm
.005	0.13
.052	1.32
.115	2.92
.320	8.13
.375	9.53
.527	13.39

Jackscrew designator L

Style	Connector size (number of contacts)	"A" Thread class 2	"B" max	C dia max	D dia max	E dia ±.002 (0.05 mm)
Male	34, 42, 50, 66, 75	.138-32 UNC	2.438 (61.93)	.192 (4.88)	.192 (4.88)	.111 (2.82)
Female	34, 42, 50, 66, 75	.138-32 UNC	2.438 (61.93)	.192 (4.88)	.192 (4.88)	.111 (2.82)
Male	104	.164-32 UNC	3.282 (83.36)	.225 (5.72)	.225 (5.72)	.140 (3.56)
Female	104	.164-32 UNC	3.282 (83.36)	.225 (5.72)	.225 (5.72)	.140 (3.56)

For use with shields 3, 4, 9, A, D, and E or plug shells S thru Z, and receptacle shells 1 thru 8.

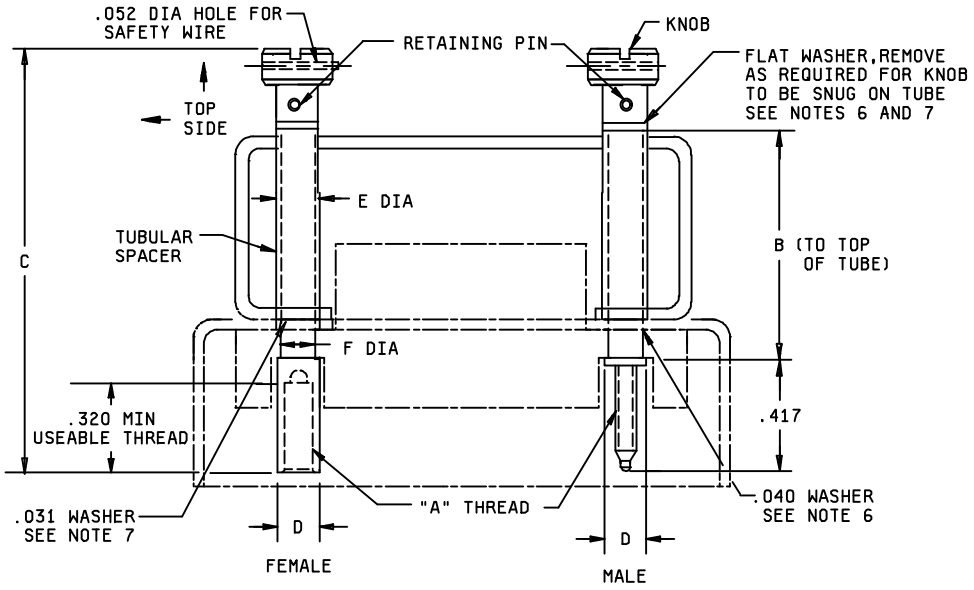
When plug shells S thru Z or receptacle polarizing shells 1 thru 8 are required to be used with a shield, use shield configurations 1, 2, B, and C.

Connector with long turnable jackscrews

Configuration L

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
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Inches	mm
.031	0.79
.040	1.02
.052	1.32
.320	8.13
.417	10.59

MULTIPIECE CONSTRUCTION

Jackscrew designator M

Style	Connector size (number of contacts)	"A" Thread class 2	"B" ±.016 (0.41 mm)	C max	D dia max	E dia ±.004 (0.10 mm)	F dia ±.003
Male	9, 14, 20, 34, 42, 50, 66, 75	6-32 UNC-2A .138-32 UNC	1.500 (43.10)	2.467 (62.66)	.190 (4.83)	.188 (4.78)	.115 (2.92)
Female	9, 14, 20, 34, 42, 50, 66, 75	6-32 UNC-2B .138-32 UNC	1.500 (43.10)	2.467 (62.66)	.190 (4.83)	.188 (4.78)	.115 (2.92)
Male	104	8-32 UNC-2A .164-32 UNC	2.495 (63.37)	3.438 (87.33)	.225 (5.72)	.219 (5.56)	.137 (3.48)
Female	104	8-32 UNC-2B .164-32 UNC	2.495 (63.37)	3.438 (87.43)	.225 (5.72)	.219 (5.56)	.137 (3.48)

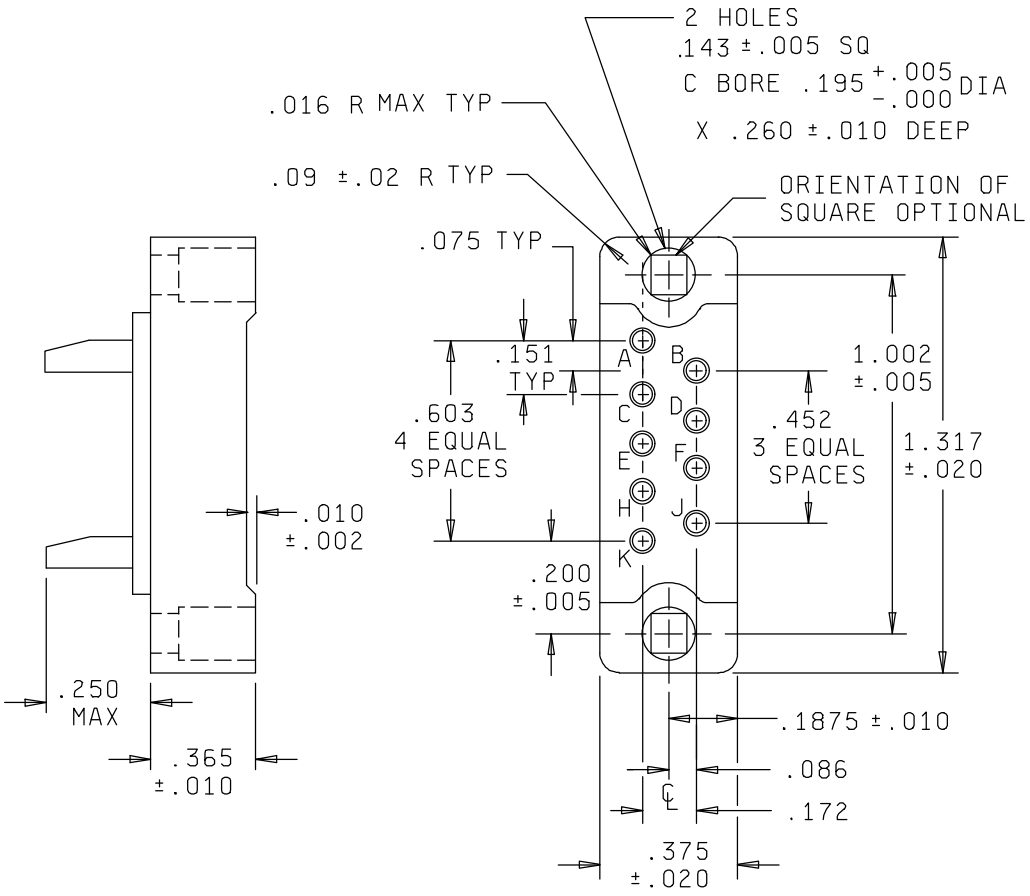
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material corrosion resistance steel in accordance with QQ-S-763 or QQ-S-764, classes 302, 303, 303se, 304 or 316 except for aluminum alloy knob and tubular, spacer, on multipiece construction.
5. Finish passivated in accordance with QQ-P-35 except for anodized knob on multipiece construction.
6. Washer .040 inch thick (1.02 mm) supplied with all sizes, not used with shell.
7. Washer .031 inch thick (0.79 mm) supplied with shield sizes 34 through 75.
8. When plug shells A through H and receptacle shells J through R are required to be used with a shield, use shield configurations 1, 2, B, and C.

Connector with long turnable jackscrews

Configuration E

FIGURE 1. Connector with shields, jackscrews, and guidepins - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
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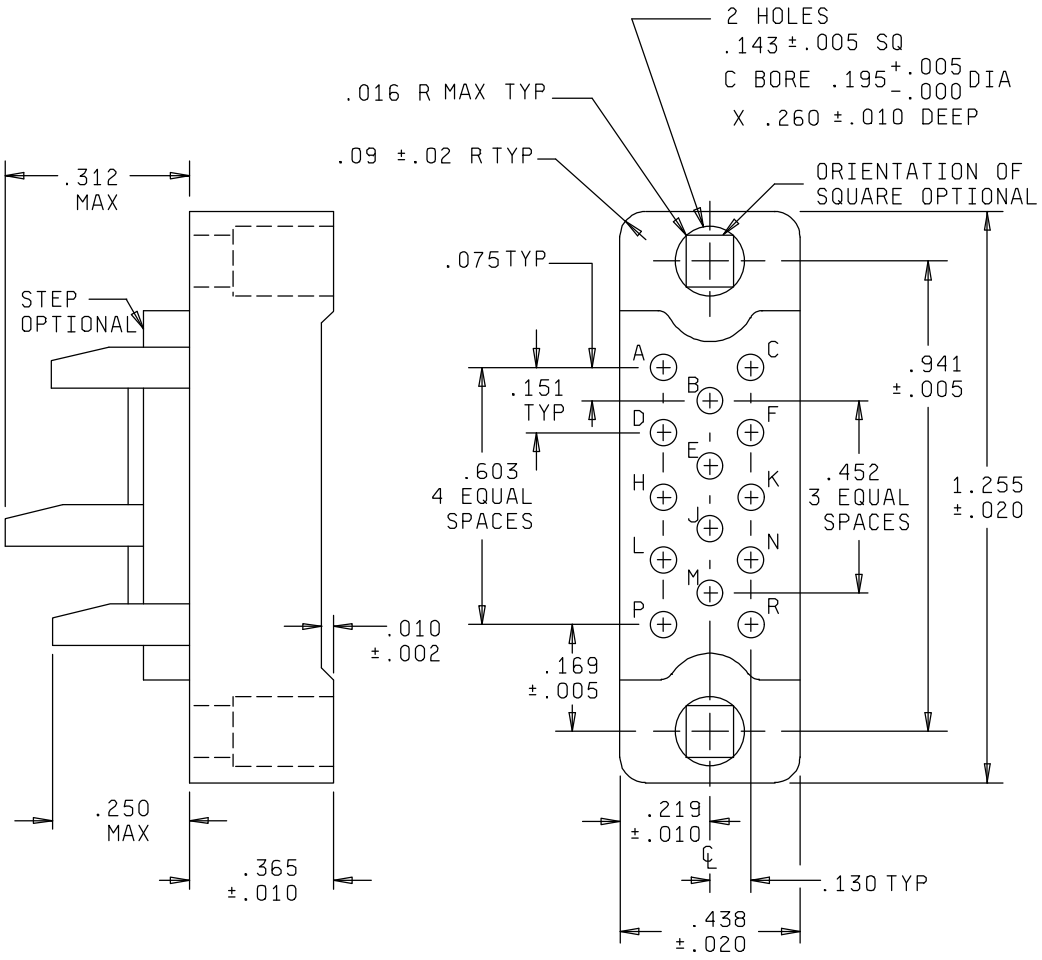
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.143	3.63	.260	6.60
.005	0.13	.151	3.84	.365	9.27
.010	0.25	.172	4.37	.375	9.53
.016	0.41	.1875	4.762	.452	11.48
.020	0.51	.195	4.95	.603	15.32
.075	1.91	.200	5.08	1.002	25.45
.086	2.18	.250	6.35	1.317	33.45

Insert with 9 contacts

Insert designators A and R

FIGURE 2. Insert arrangements.

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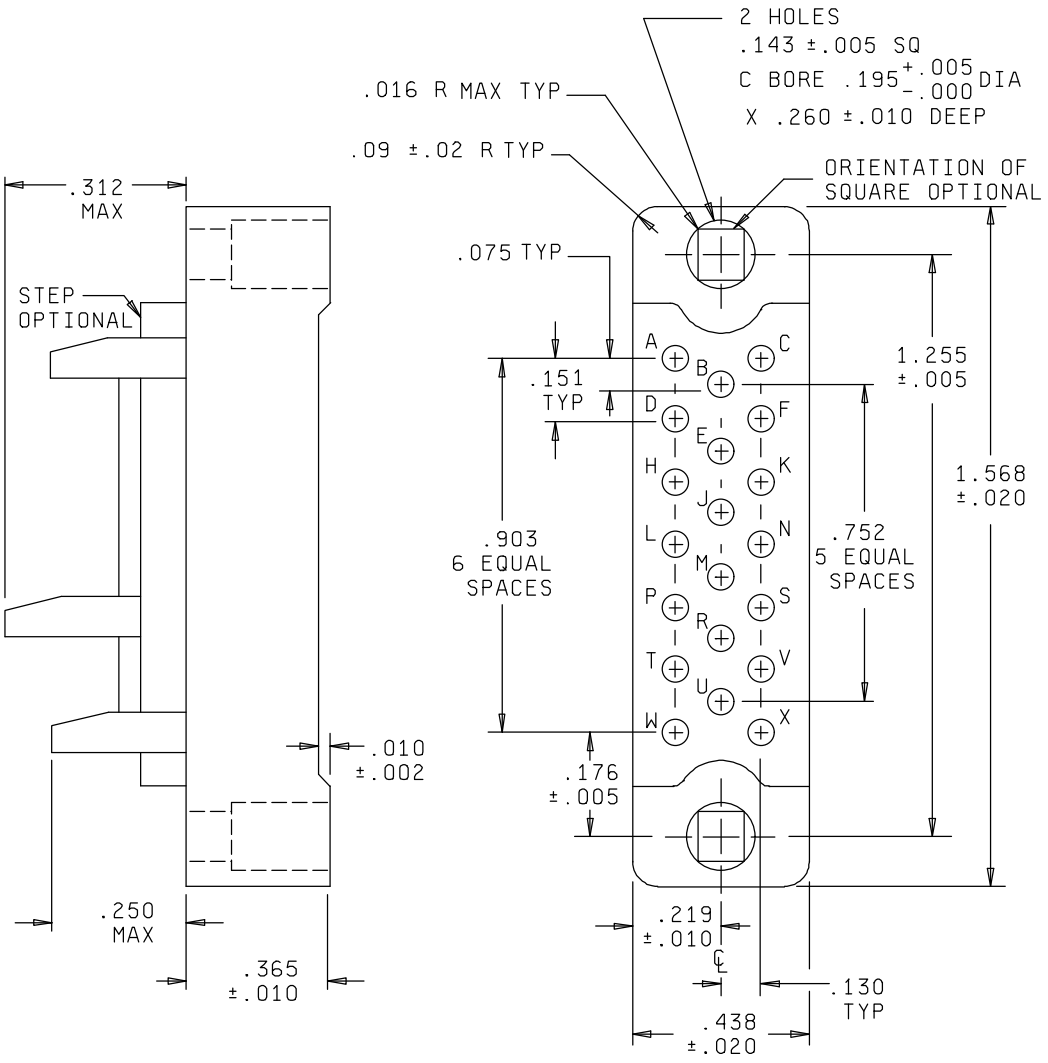
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.130	3.30	.312	7.92
.005	0.13	.143	3.63	.365	9.27
.010	0.25	.151	3.84	.438	11.13
.016	0.41	.169	4.29	.452	11.48
.02	0.5	.195	4.95	.603	15.32
.020	0.51	.219	5.56	.941	23.90
.075	1.91	.250	6.35	1.255	31.88
.09	2.3	.260	6.60		

Insert with 14 contacts

Insert designators B and S

FIGURE 2. Insert arrangements - Continued.

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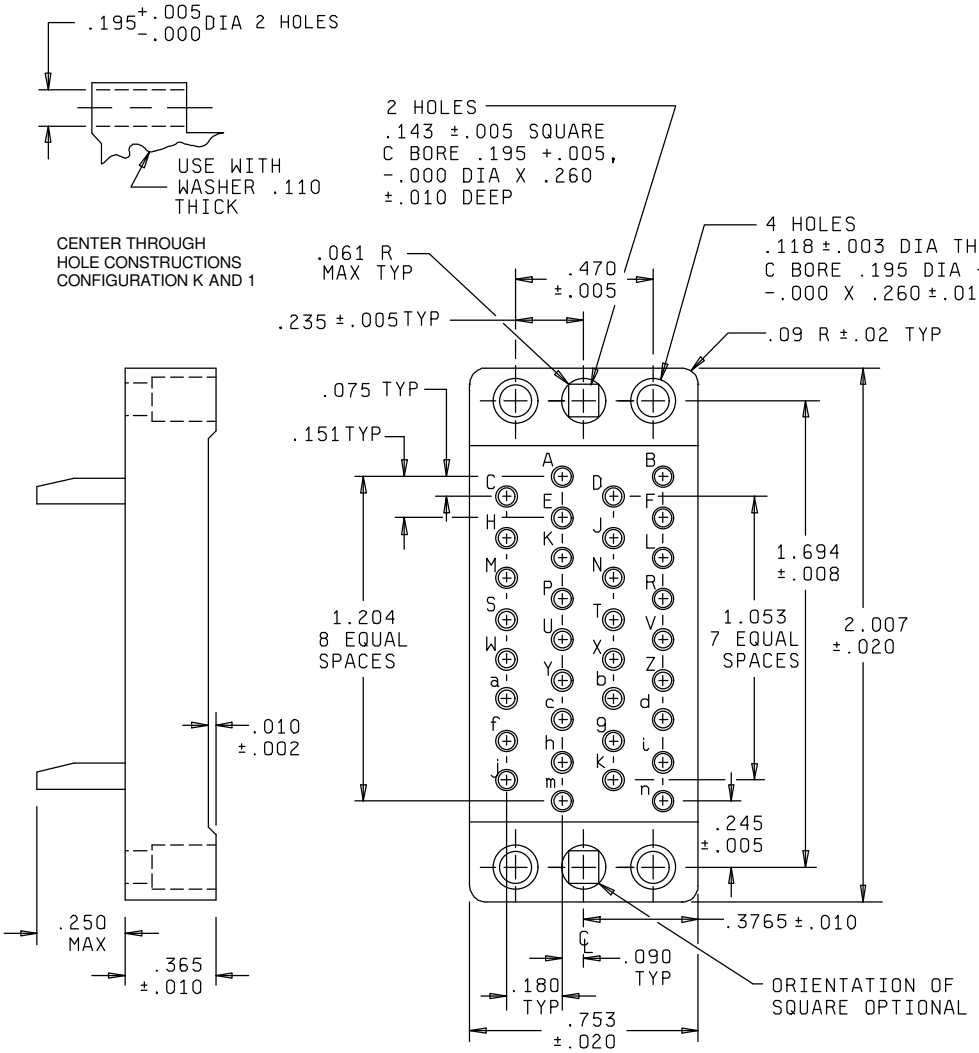
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.130	3.30	.312	7.92
.005	0.13	.143	3.63	.365	9.27
.010	0.25	.151	3.84	.438	11.13
.016	0.41	.176	4.47	.752	19.10
.02	0.5	.195	4.95	.903	22.94
.020	0.51	.219	5.56	1.255	31.88
.075	1.91	.250	6.35	1.568	39.89
.09	2.3	.260	6.60		

Insert with 20 contacts

Insert designators C and T

FIGURE 2. Insert arrangements - Continued.

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Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.061	1.55
.075	1.91
.09	2.3
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.151	3.84
.180	4.57
.195	4.95
.235	5.97
.245	6.22
.250	6.35
.260	6.60
.365	9.55
.3765	9.563
.470	11.94
.753	19.13
1.053	26.75
1.204	30.58
1.694	43.03
2.007	50.98

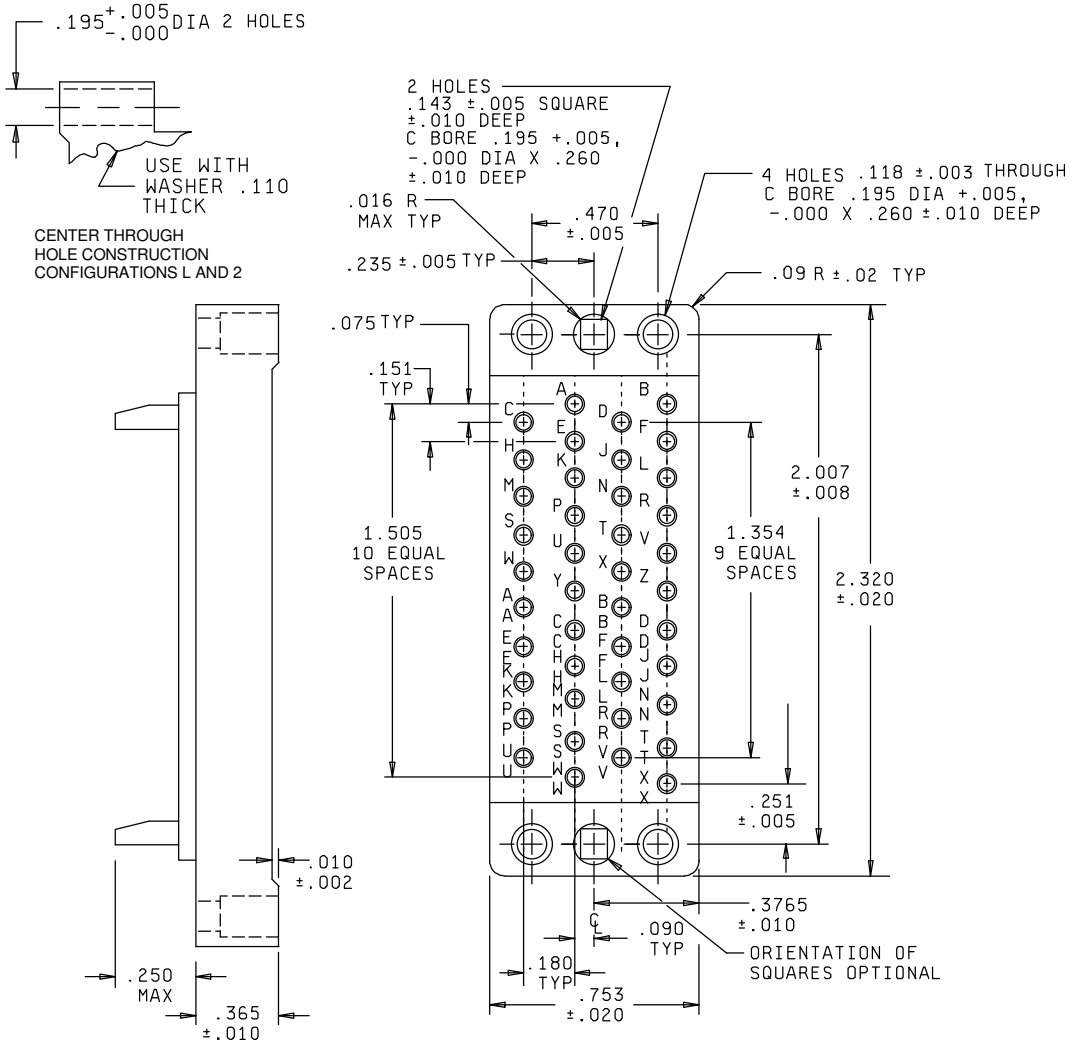
Configurations D and U - Suitable for all applications, except one-piece jackscrews.
Configurations K and 1 - For one piece jackscrews (through hole construction).

Insert with 34 contacts

Insert designators D, K, U, and 1

FIGURE 2. Insert arrangements - Continued.

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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.110	2.79	.260	6.60
.003	0.08	.118	3.00	.365	9.27
.005	0.13	.143	3.63	.3765	9.56
.010	0.25	.151	3.84	.470	11.94
.016	0.41	.180	4.57	.753	19.13
.020	0.51	.195	4.95	1.354	34.39
.075	1.91	.235	5.97	1.505	38.23
.09	2.29	.250	6.35	2.007	50.98
.090	2.29	.251	6.38	2.320	58.93

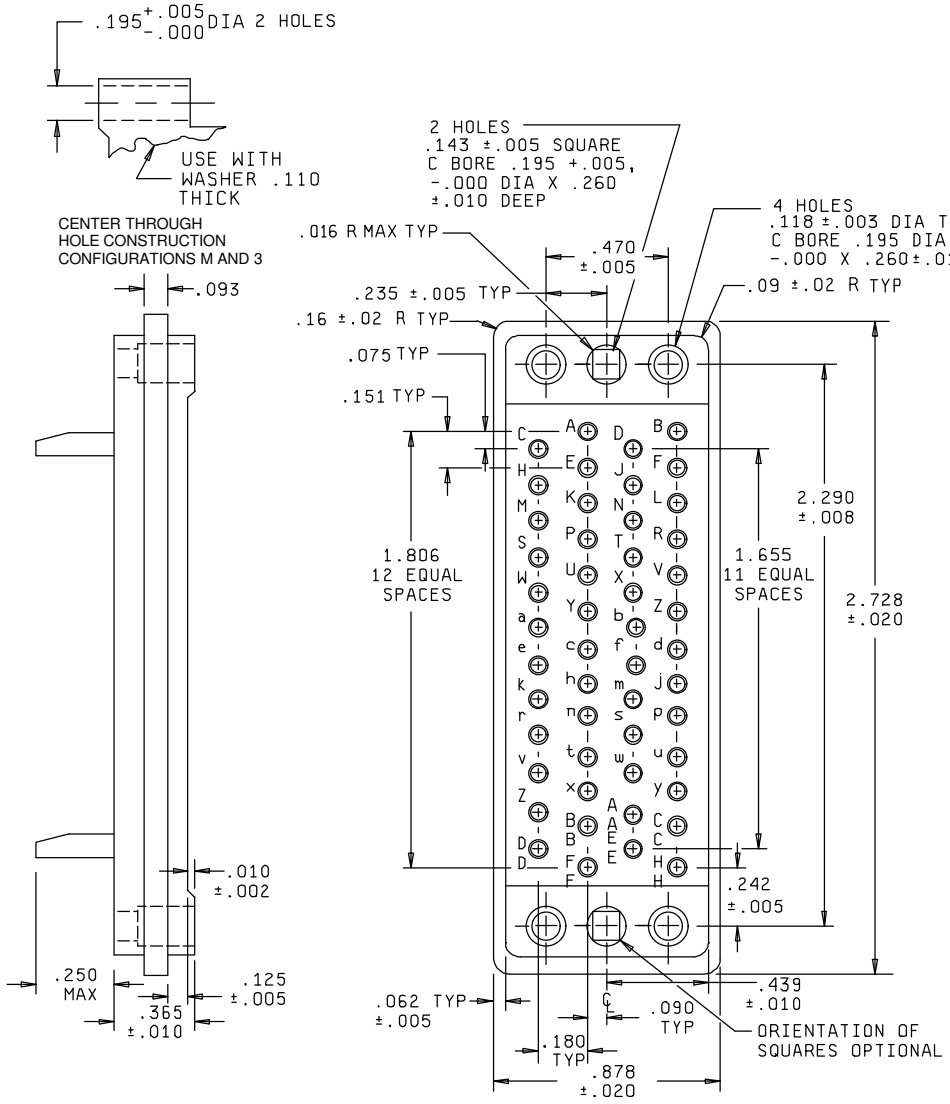
Configurations E and V - Suitable for all applications, except one-piece jackscrews.
Configurations L and 2 - For one-piece jackscrews (through hole construction).

Insert with 42 contacts

Insert designators E, L, V, and 2

FIGURE 2. Insert arrangements - Continued.

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Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.02	0.5
.020	0.51
.062	1.57
.075	1.91
.09	2.29
.090	2.29
.093	2.36
.110	2.79
.118	3.00
.125	3.18
.143	3.63
.151	3.84
.16	4.06
.180	4.57
.195	4.95
.235	5.97
.242	6.15
.250	6.35
.260	6.60
.365	9.27
.439	11.15
.470	31.88
.878	23.30
1.655	42.04
1.806	45.87
2.290	58.17
2.728	69.29

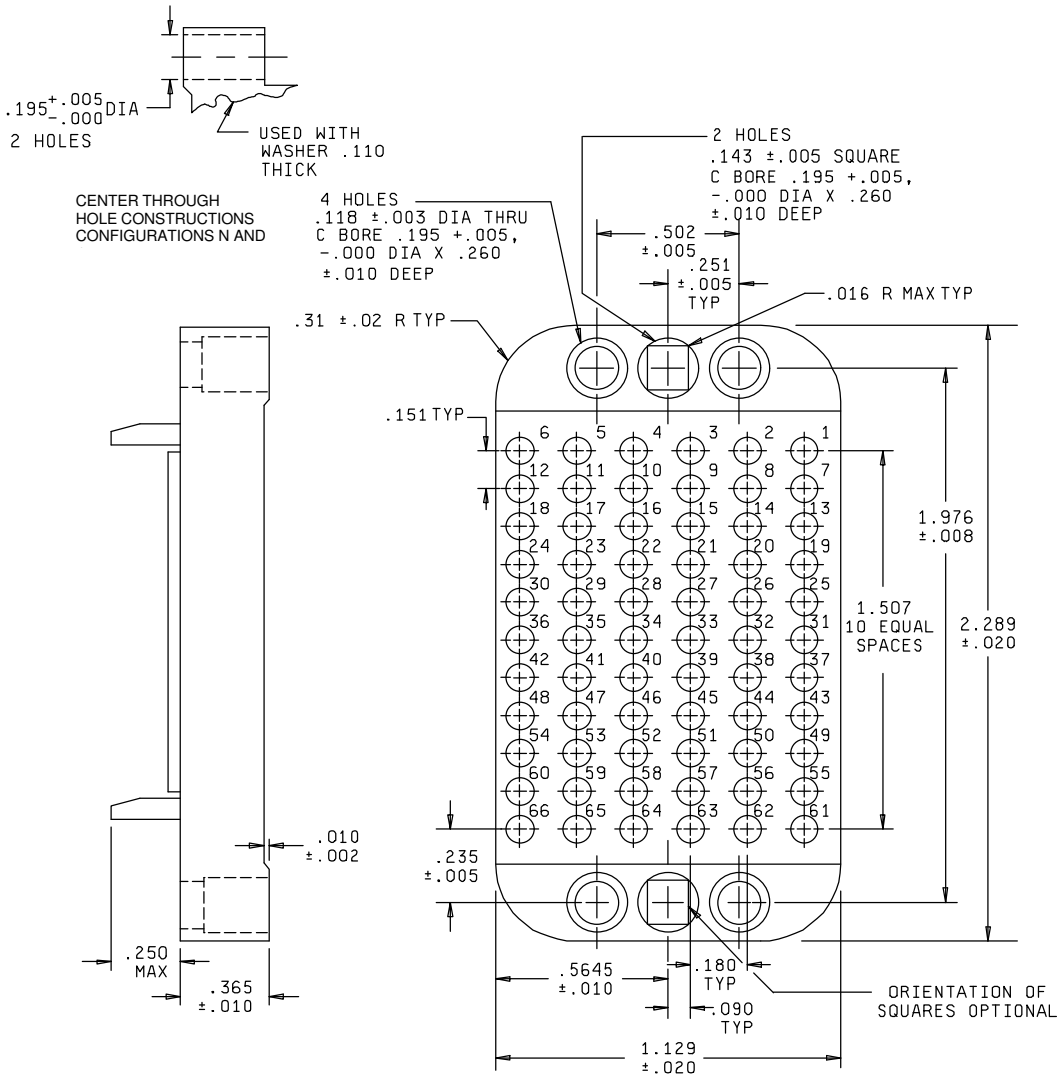
Configurations F and W - Suitable for all applications, one-piece jackscrews.
Configurations M and 3 - For one-piece jackscrews (through hole construction).

Insert with 50 contacts

Insert designators F, M, W, and 3

FIGURE 2. Insert arrangements - Continued.

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Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.5
.020	0.51
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.151	3.84
.156	3.96
.180	4.57
.195	4.95
.235	5.97
.250	6.35
.251	6.38
.260	6.60
.31	7.87
.365	9.72
.502	12.75
.5645	14.34
1.129	28.68
1.507	38.28
1.976	50.19
2.289	58.14

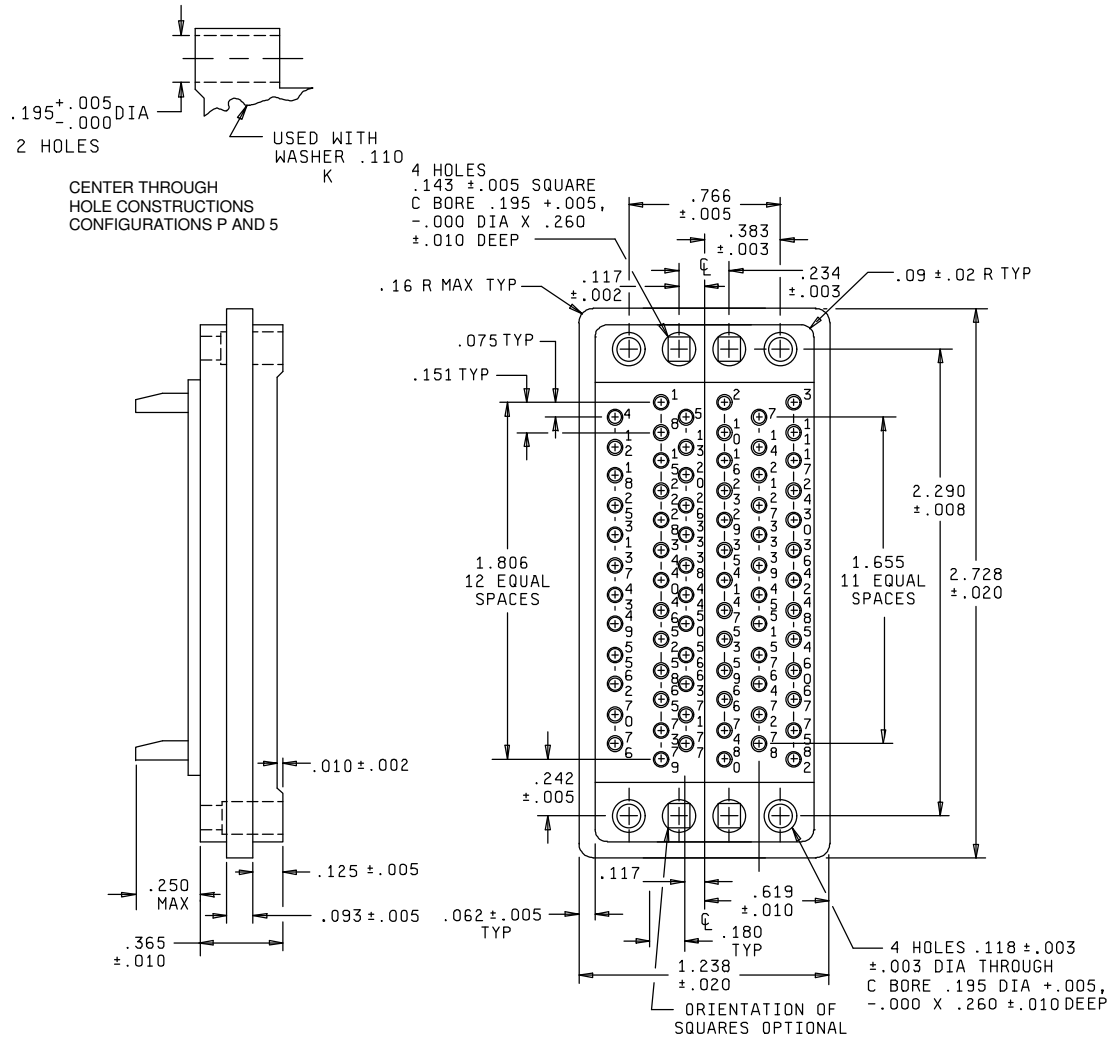
Configurations G and X - Suitable for all applications, except one-piece jackscrews.
Configurations N and 4 - one-piece jackscrews (through hole construction).

Insert with 66 contacts

Insert designators G, N, X, and 4

FIGURE 2. Insert arrangements - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.09	2.3	.180	4.57	.384	9.75
.003	0.08	.093	2.36	.195	4.95	.619	12.75
.005	0.13	.110	2.79	.234	5.97	.766	19.46
.010	0.25	.117	2.97	.242	6.15	.769	19.53
.016	0.41	.118	3.00	.250	6.35	1.238	31.45
.02	0.5	.125	3.18	.260	6.60	1.655	42.04
.020	0.51	.143	3.63	.31	7.87	1.806	44.97
.062	1.57	.151	3.84	.365	9.27	2.290	58.17
.075	1.91	.16	4.1	.383	9.73	2.728	59.29

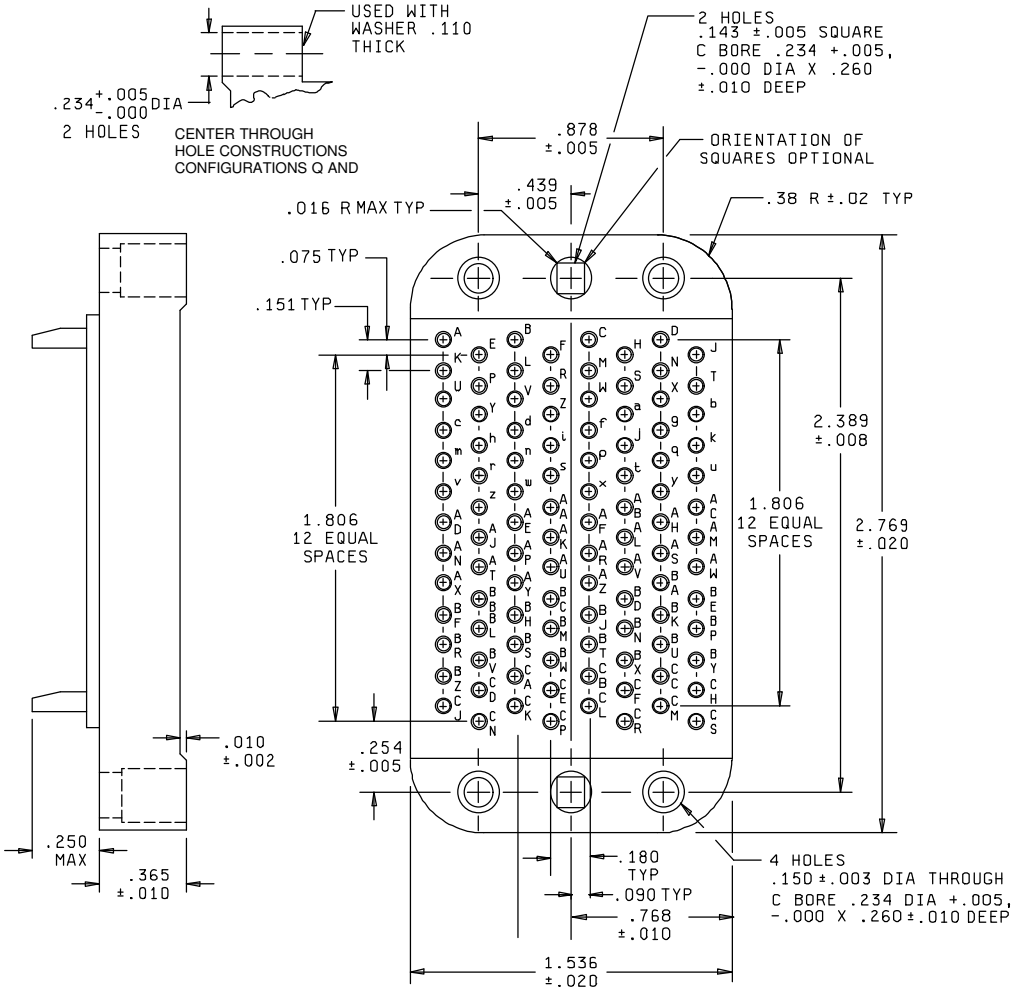
Configurations H and Y - Suitable for all applications, except one-piece jackscrews.
Configurations P and 5 - For one-piece jackscrews (through hole construction).

Insert with 75 contacts

Insert designators H, P, Y, and 5

FIGURE 2. Insert arrangements - Continued.

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Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm
.002	0.05	.090	2.29	.260	6.60
.003	0.08	.110	2.79	.365	9.27
.005	0.13	.143	3.63	.38	9.65
.008	0.20	.150	3.81	.439	11.15
.010	0.25	.151	3.84	.768	19.51
.016	0.41	.180	4.57	.878	22.30
.020	0.51	.234	5.94	1.536	39.01
.062	1.57	.250	6.35	1.806	45.87
.075	1.91	.254	6.45	2.389	60.68
				2.769	70.33

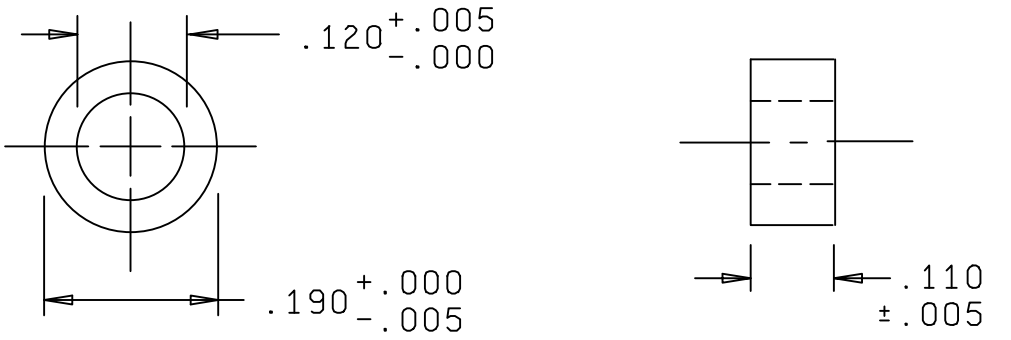
Configurations J and Z - Suitable for all applications, except one-piece jackscrews.
Configurations Q and 6 - For one-piece jackscrews (through hole construction).

Insert with 104 contacts

Insert designators J, Q, Z, and 6

FIGURE 2. Insert arrangements - Continued.

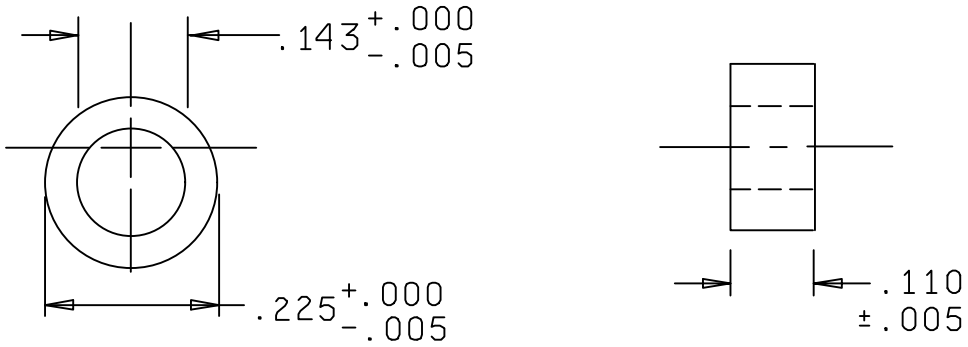
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
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Inches	mm
.005	0.13
.110	2.79
.120	3.05
.190	4.83

(34, 42, 50, and 75 contacts)

For use with inserts K, L, M, N, P (through hole).



(104 contacts)

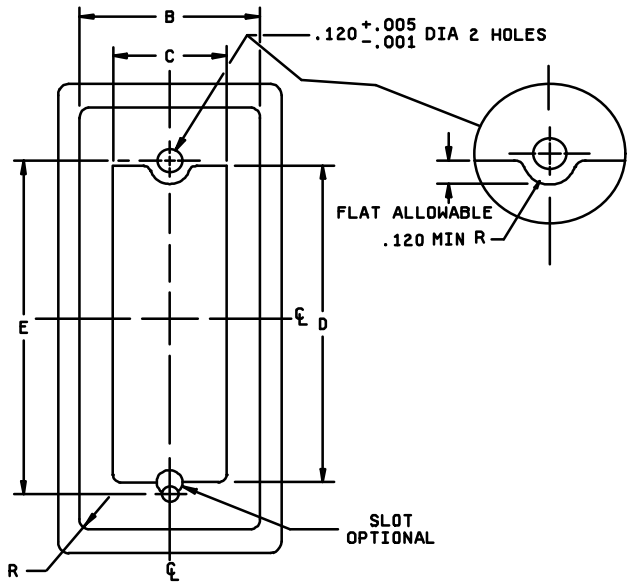
For use with Insert Q (through hole).

NOTES:

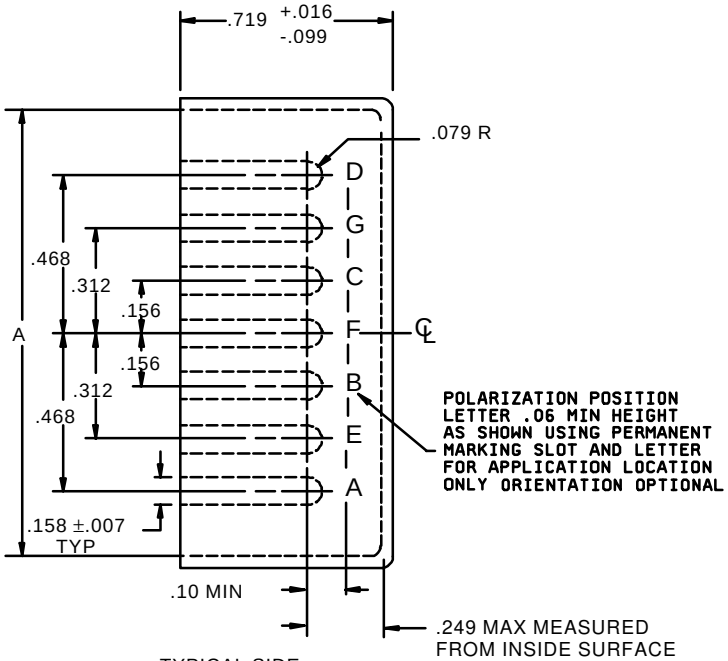
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.015 inch (0.38 mm) for three place decimals.
4. Tolerance between any two adjacent contact centers shall be ±.004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent contacts, shall be ±.006 inch (0.15 mm).
5. Contact identification may be located on either side of the contact hold but shall appear on the front and rear face of the contact insert.
6. Designed for use with MIL-DTL-28748 size 16 contacts.
7. Insert material shall be in accordance with MIL-DTL-28748.
8. Washer and self locking nut: Corrosion resistant steel, in accordance with QQ-S-763 or QQ-S-764, classes 302, 303, 303Se, 304, or 316.
9. Washer finish: Passivated in accordance with QQ-P-35 and self locking nut.

FIGURE 2. Insert arrangements - Continued.

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STYLE FOR 9, 14, AND 20
CONTACT CONNECTORS
(J THRU R)

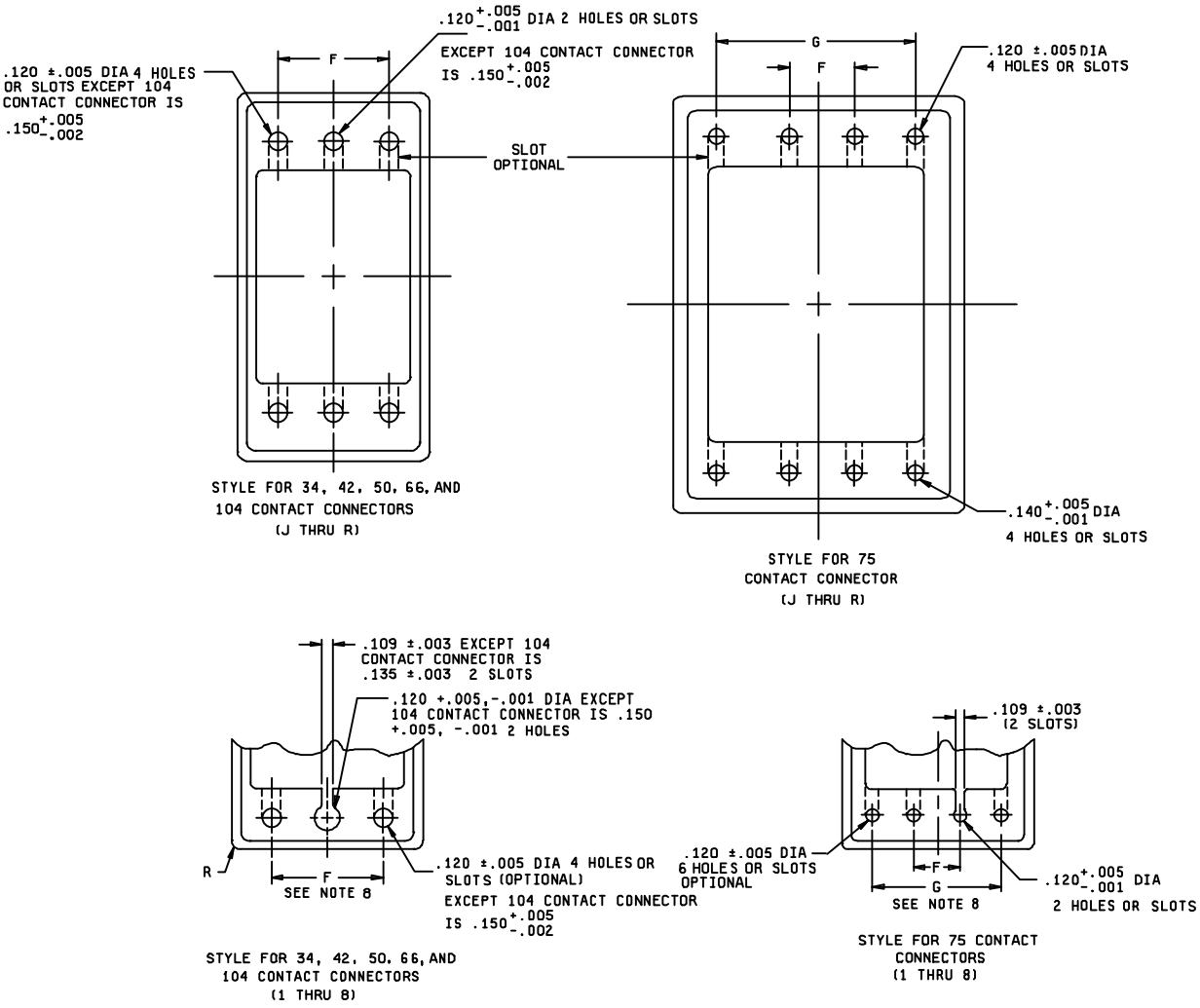


TYPICAL SIDE
VIEW FOR ALL
RECEPTACLE SHELLS

Receptacle shell polarizing

FIGURE 3. Shell configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
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Inches	mm	Inches	mm
.001	0.03	.120	3.05
.002	0.05	.125	3.18
.003	0.08	.135	3.43
.005	0.13	.140	3.56
.007	0.18	.150	3.81
.016	0.41	.156	3.96
.021	0.53	.158	4.01
.062	1.57	.234	5.94
.063	1.60	.249	6.32
.070	1.78	.312	7.92
.079	2.01	.452	11.48
.099	2.51	.468	11.89
.10	2.54	.719	18.26
.109	2.77	.936	23.77
.116	2.95		

Receptacle shell polarizing

FIGURE 3. Shell configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.

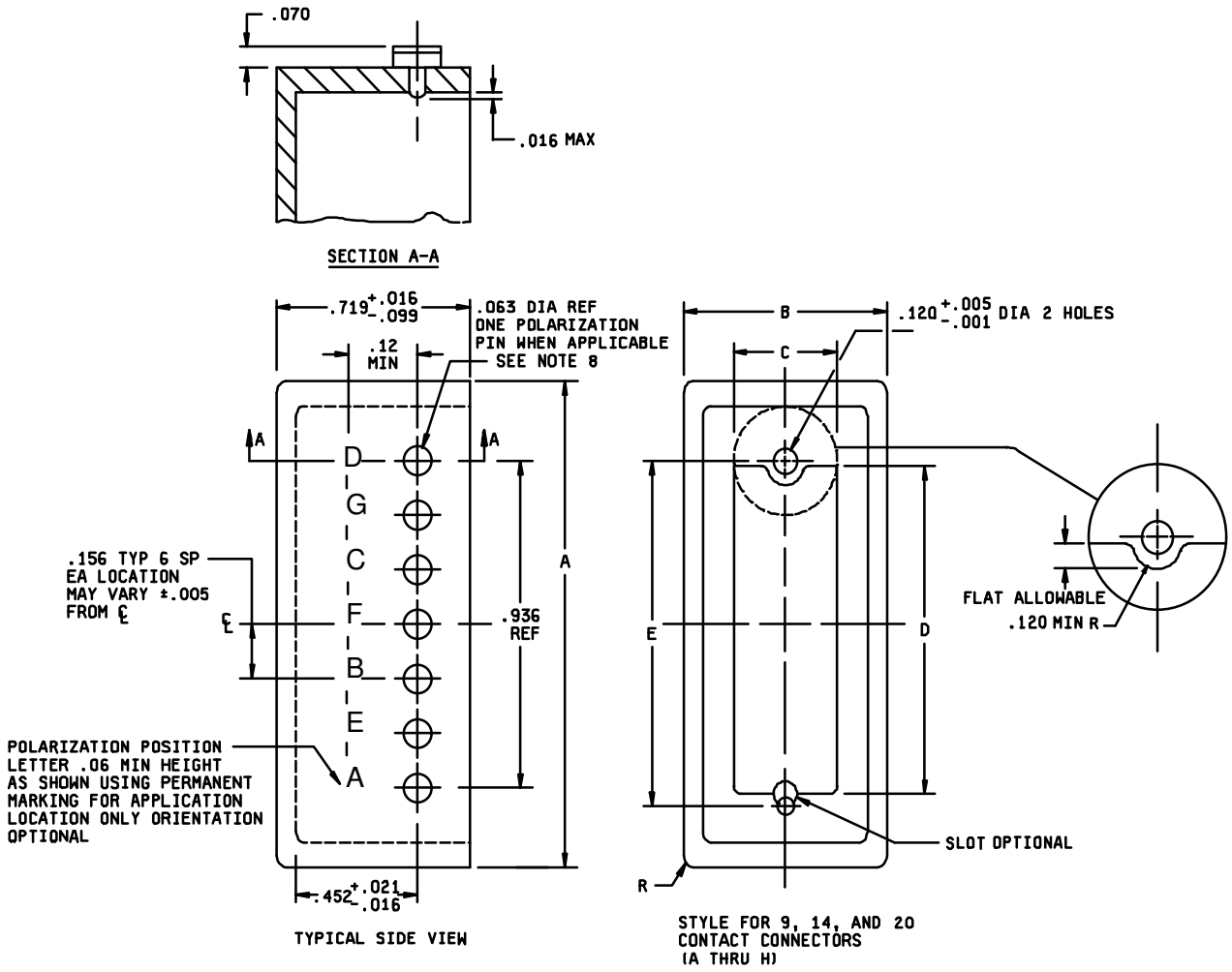
Polarization designators	Connector size (number of contacts)	A min	B min	C min	D min	E ±.005 (0.13 mm)	F ±.005 (0.13 mm)	G ±.005 (0.13 mm)	R
J thru R	9	1.448 (36.78)	.512 (13.00)	.375 (9.52)	.864 (21.94)	1.000 (25.40)	---		.141 (3.58)
J thru R	14	1.388 (35.36)	.575 (14.60)	.440 (11.18)	.800 (20.32)	.938 (23.82)	---		.141 (3.58)
J thru R	20	1.702 (43.23)	.575 (14.60)	.440 (11.18)	1.097 (27.86)	1.250 (31.75)	---		.141 (3.58)
J thru R	34	2.138 (54.30)	.888 (22.56)	.745 (18.92)	1.400 (35.76)	1.688 (42.88)	.468 (11.89)		.141 (3.58)
J thru R	42	2.450 (62.23)	.888 (22.56)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)		.141 (3.58)
J thru R	50	2.856 (72.54)	1.015 (25.78)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)		.188 4/ (4.78)
J thru R	66	2.420 (61.47)	1.264 (32.11)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01	.500 (12.70)		.375 (9.52)
J thru R	75	2.855 (72.52)	1.370 (34.80)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.203 (5.16)
J thru R	104	2.900 (73.66)	1.680 (42.67)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)		.422 (10.72)
1 thru 8 (see note 8)	34	2.138 (54.30)	.888 (22.56)	.745 (18.92)	1.400 (35.56)	1.688 (42.88)	.468 (11.89)		.141 (3.58)
1 thru 8 (see note 8)	42	2.450 (62.23)	.888 (22.56)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)		.141 (3.58)
1 thru 8 (see note 8)	50	2.856 (72.54)	1.015 (25.78)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)		.188 4/ (4.78)
1 thru 8 (see note 8)	66	2.420 (61.47)	1.264 (32.10)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01)	.500 (12.70)		.375 (9.52)
1 thru 8 (see note 8)	75	2.855 (72.52)	1.370 (34.80)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.203 (5.16)
1 thru 8 (see note 8)	104	2.900 (73.66)	1.680 (42.67)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)		.422 (10.72)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Tolerance +.020 inch (0.51 mm) -.016 inch (0.41 mm).
5. Material aluminum alloy in accordance with specification QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
6. Finish anodize in accordance with MIL-A-8625, type I or type II.
7. Each slot may vary ±.005 inch (0.13 mm) from centerline.
8. Receptacle shell designators 1, 2, 3, 4, 5, 6, 7, and 8 are not used with one-piece jackscrew shield designators 3, 4, 9, A, D, and E.

Receptacle shell polarizing

FIGURE 3. Shell configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.

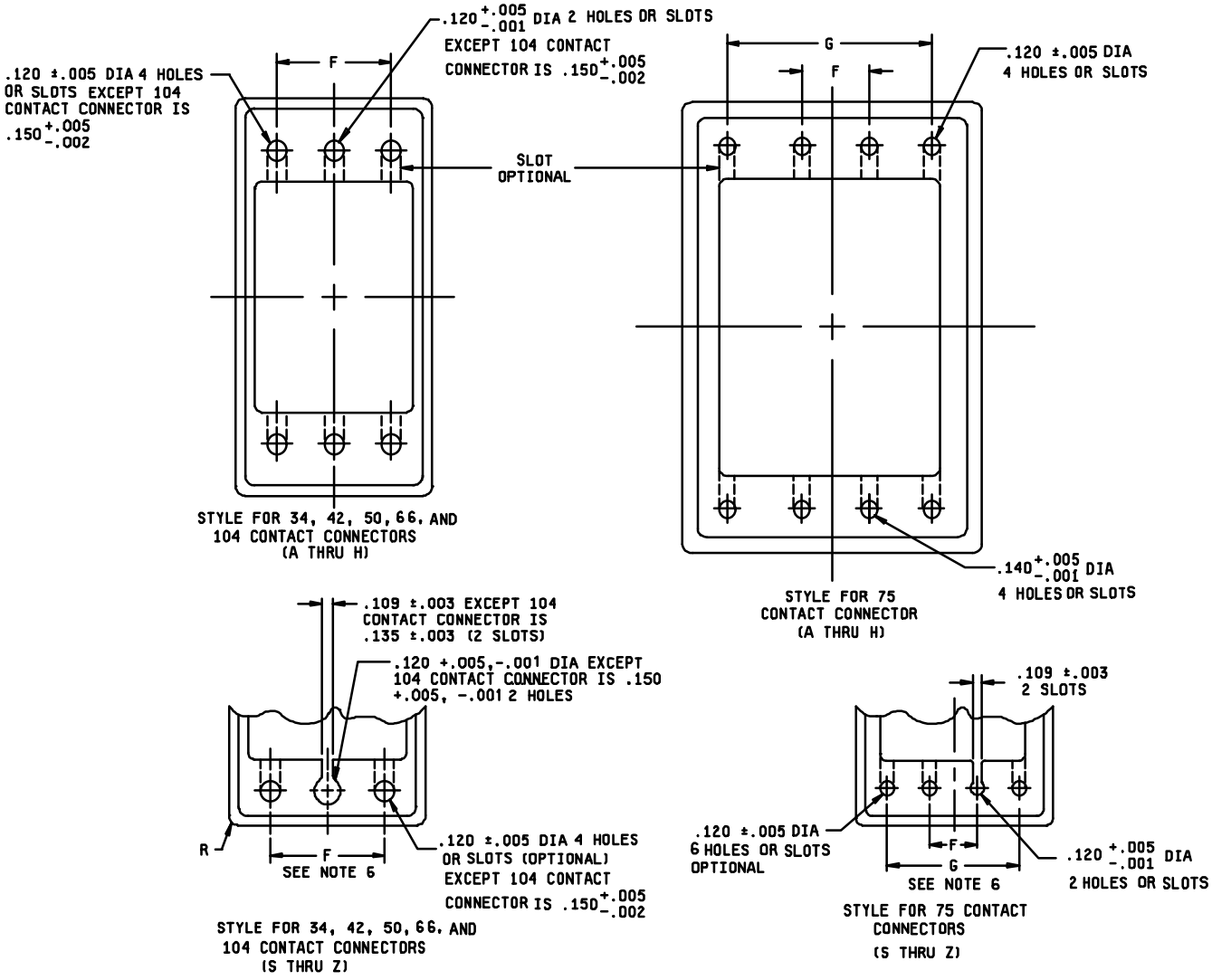


Inches	mm	Inches	mm
.001	0.03	.099	2.51
.005	0.13	.116	2.95
.016	0.41	.120	3.05
.021	0.53	.156	3.96
.06	1.52	.452	11.48
.070	1.78	.719	18.26

Plug shell polarizing

FIGURE 3. Shell configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm
.001	0.03	.116	2.95
.002	0.05	.120	3.05
.003	0.08	.125	3.18
.005	0.13	.135	3.43
.016	0.41	.140	3.56
.021	0.53	.150	3.81
.062	1.57	.156	3.96
.063	1.60	.234	5.94
.070	1.78	.452	11.48
.099	2.51	.719	18.26
.109	2.77	.936	23.77

FIGURE 3. Shell configurations - Continued.

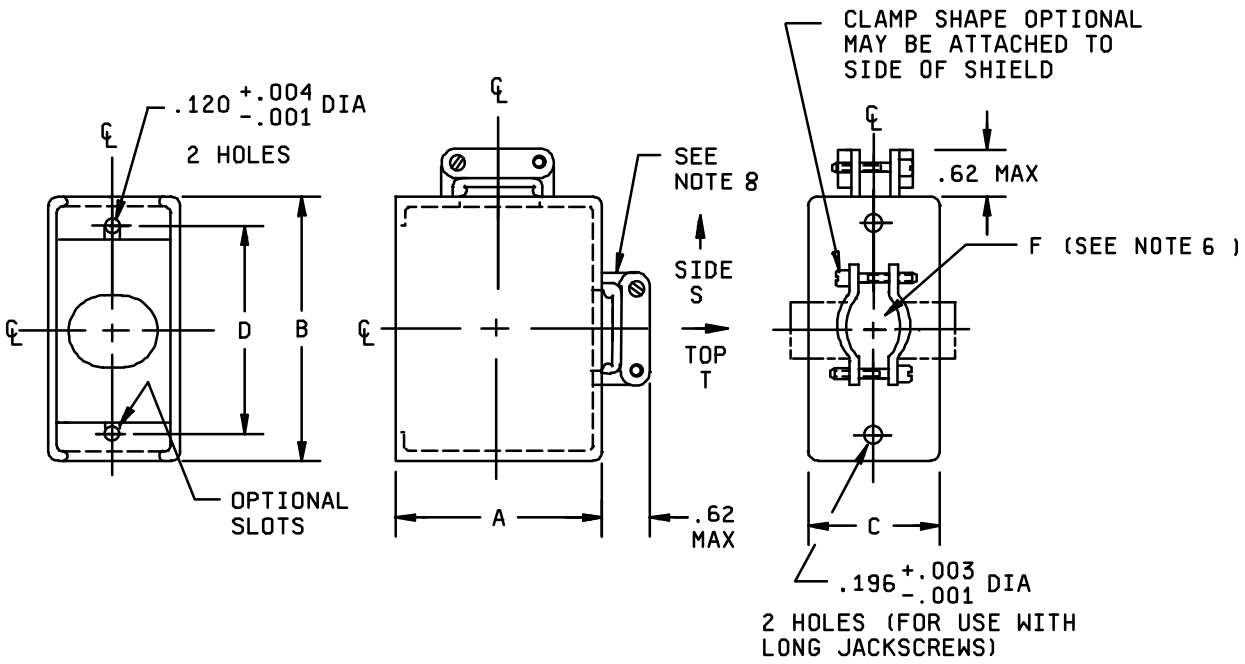
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.

Shell suffix designator	Connector size (number of contacts)	A max	B max	C min	D min	E ±.005 (0.13 mm)	F ±.005 (0.13 mm)	G ±.005 (0.13 mm)	R
A thru H	9	1.440 (36.58)	.500 (12.70)	.375 (9.52)	.864 (21.94)	1.000 (25.40)	---		.141 (3.58)
A thru H	14	1.380 (35.05)	.567 (14.40)	.440 (11.18)	.800 (20.32)	.938 (23.82)	---		.141 (3.58)
A thru H	20	1.700 (43.18)	.567 (14.40)	.440 (11.18)	1.097 (27.86)	1.250 (31.75)	---		.141 (3.58)
A thru H	34	2.130 (54.10)	.880 (22.35)	.745 (18.92)	1.400 (35.76)	1.688 (42.88)	.468 (11.89)		.141 (3.58)
A thru H	42	2.442 (62.03)	.880 (22.35)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)		.141 (3.58)
A thru H	50	2.846 (72.39)	1.005 (25.53)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)		.188 (4.78)
A thru H	66	2.410 (61.21)	1.254 (31.85)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01	.500 (12.70)		.375 (9.52)
A thru H	75	2.850 (72.39)	1.365 (34.67)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.187 (4.75)
A thru H	104	2.890 (73.41)	1.670 (42.42)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)		.422 (10.72)
S thru Z (see note 6)	34	2.130 (54.10)	.880 (22.35)	.745 (18.92)	1.400 (35.56)	1.688 (42.88)	.468 (11.89)		.141 (3.58)
S thru Z (see note 6)	42	2.442 (62.03)	.880 (22.35)	.745 (18.92)	1.700 (43.18)	2.000 (50.80)	.468 (11.89)		.141 (3.58)
S thru Z (see note 6)	50	2.846 (72.29)	1.005 (25.53)	.745 (18.92)	2.026 (51.46)	2.281 (57.94)	.468 (11.89)		.188 (4.78)
S thru Z (see note 6)	66	2.410 (61.21)	1.254 (31.85)	1.115 (28.32)	1.696 (43.08)	1.969 (50.01)	.500 (12.70)		.375 (9.52)
S thru Z (see note 6)	75	2.850 (72.39)	1.365 (34.67)	1.104 (28.04)	2.026 (51.46)	2.281 (57.94)	.234 (5.94)	.766 (19.46)	.187 (4.75)
S thru Z (see note 6)	104	2.890 (73.41)	1.670 (42.42)	1.476 (37.49)	2.073 (52.65)	2.375 (60.32)	.875 (22.22)		.422 (10.72)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.
6. Plug shell designators S, T, U, V, W, X, Y, and Z are not used with one-piece jackscrew shield designators 3,4, 9, A, D, and E.
7. One polarization pin is used when applicable.

FIGURE 3. Shell configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



Inches	mm
.001	0.03
.003	0.08
.004	0.10
.120	3.05
.196	4.98
.62	15.75

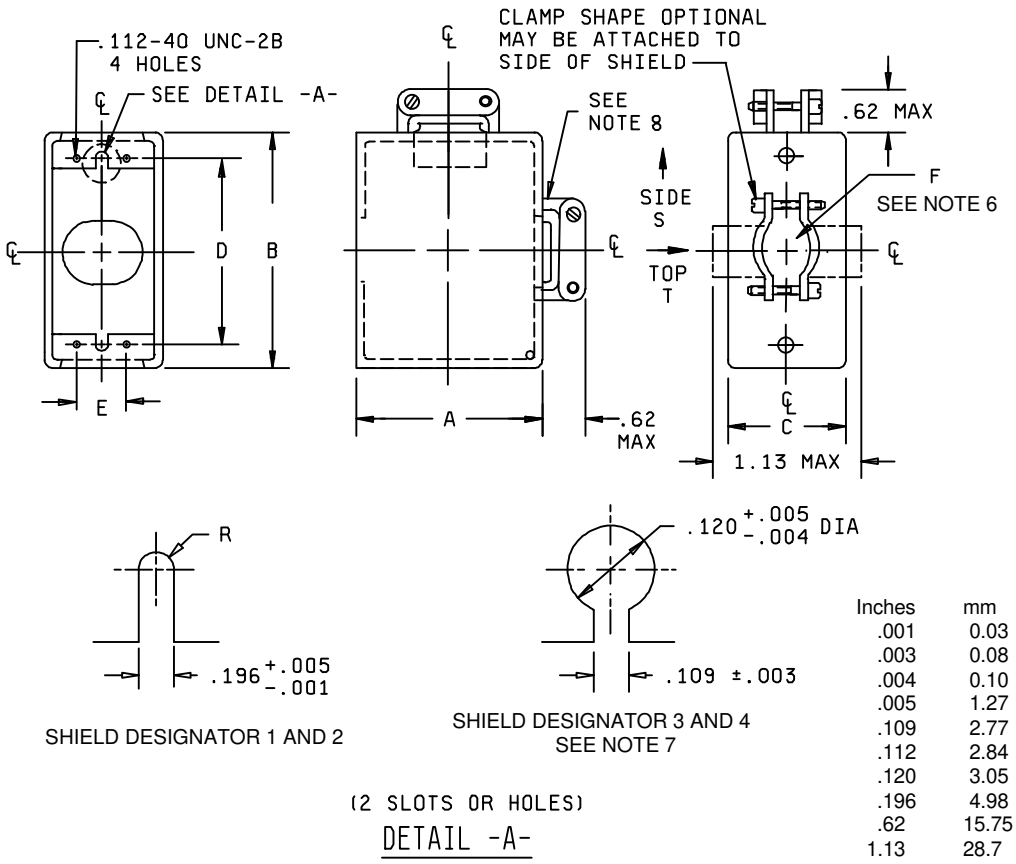
Configurations 1 and 2 are used with shell for “L” hardware and can be used with or without shell for “M” hardware.
Configurations 3 and 4 are used without shell.

Shield clamp location suffix designator	Connector size (no. of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	F Min. Cable opening sq. inch
1 - top opening 2 - side opening	9	1.18 (29.97) 1.35 (34.29)	1.312 (33.32)	.437 (11.10)	1.000 (25.40)	.10 (2.54)
1 - top opening 2 - side opening	14	1.18 (29.97) 1.35 (34.29)	1.250 (31.75)	.500 (12.70)	.938 (23.83)	.19 (4.83)
1 - top opening 2 - side opening	20	1.18 (29.97) 1.35 (34.29)	1.562 (39.67)	.500 (12.70)	1.250 (31.75)	.22 (5.59)

See notes at end of figure.

FIGURE 4. Shield.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



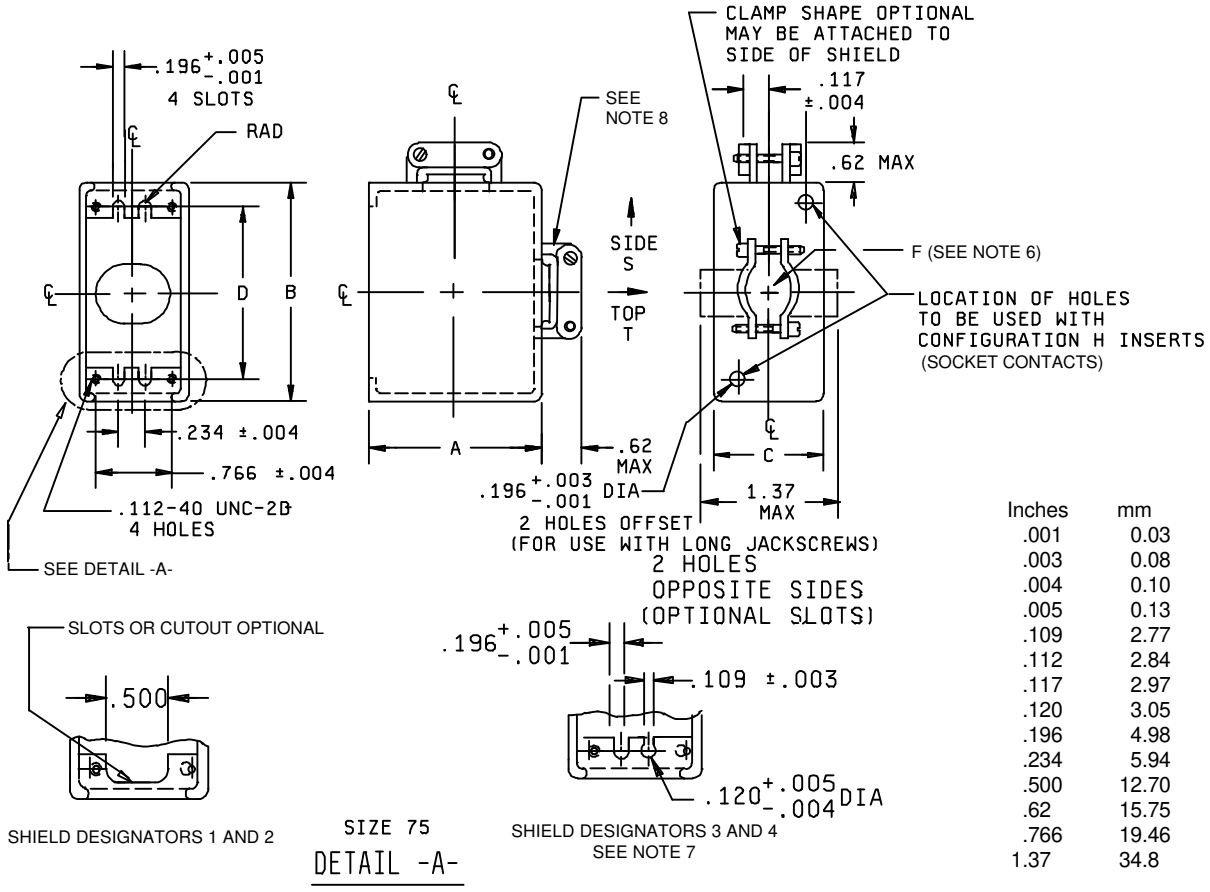
Inches	mm
.001	0.03
.003	0.08
.004	0.10
.005	1.27
.109	2.77
.112	2.84
.120	3.05
.196	4.98
.62	15.75
1.13	28.7

Configurations 1 and 2 are used with shell for “L” hardware and can be used with or without shell for “M” hardware. Configurations 3 and 4 are used without shell.

Shield clamp location suffix designator	Connector size (no. of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	E ±.010 (0.25)	F Min. Cable opening sq. inch
1 - top opening 2 - side opening	34	1.22 (30.99) 1.35 (34.29)	2.000 (50.80)	.828 (21.03)	1.688 (42.88)	.468 (11.89)	.34 (8.64)
1 - top opening 2 - side opening	42	1.22 (30.99) 1.35 (34.29)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.468 (11.89)	.34 (8.64)
1 - top opening 2 - side opening	50	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	.828 (21.03)	2.282 (57.96)	.468 (11.89)	.34 (8.64)
3 - top opening (see note 9) 4 - side opening	34	1.22 (30.99) 1.35 (34.29)	2.000 (50.80)	.828 (21.03)	1.688 (42.88)	.468 (11.89)	.34 (8.64)
3 - top opening (see note 9) 4 - side opening	42	1.22 (30.99) 1.35 (34.29)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.468 (11.89)	.34 (8.64)
3 - top opening (see note 9) 4 - side opening	50	1.22 (30.99) 1.55 (39.37)	2.594 (65.89)	.828 (21.03)	2.282 (57.96)	.468 (11.89)	.34 (8.64)

FIGURE 4. Shield - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



Inches	mm
.001	0.03
.003	0.08
.004	0.10
.005	0.13
.109	2.77
.112	2.84
.117	2.97
.120	3.05
.196	4.98
.234	5.94
.500	12.70
.62	15.75
.766	19.46
1.37	34.8

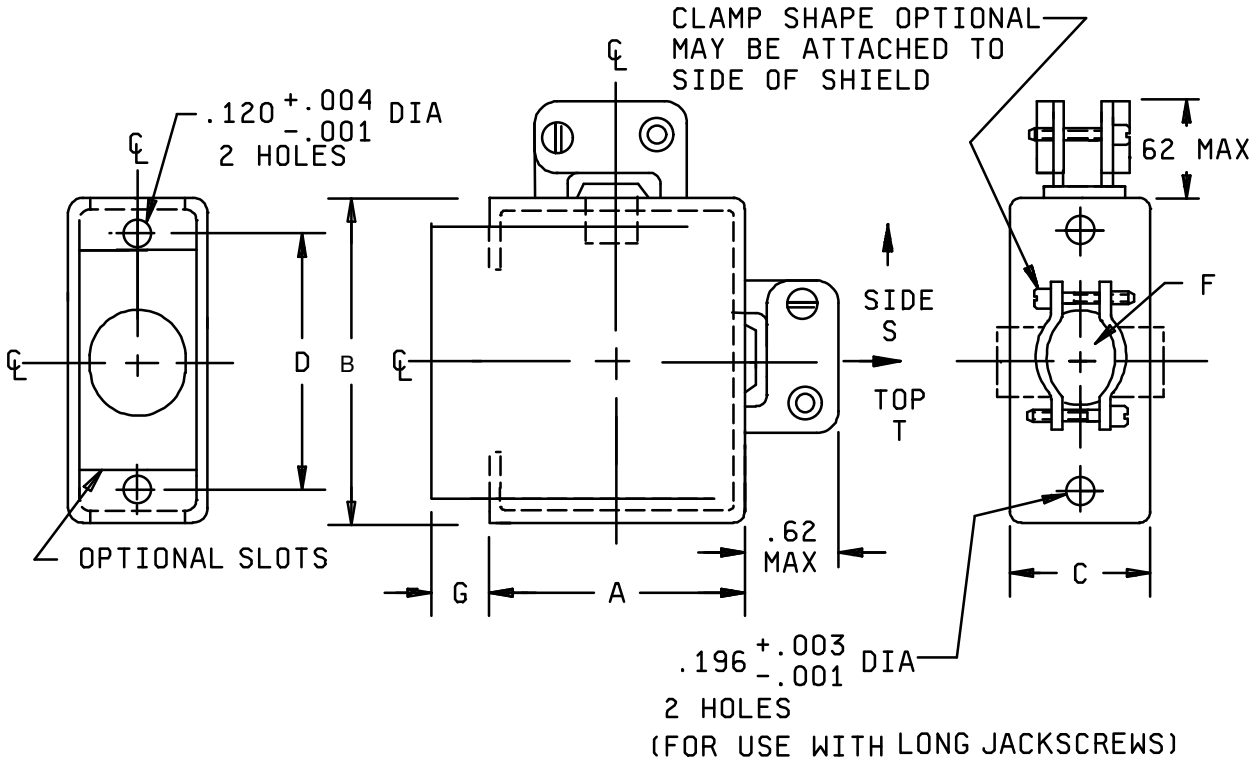
Configurations 1 and 2 are used with shell for “L” hardware and can be used with or without shell for “M” hardware.
Configurations 3 and 4 are used without shell.

Shield clamp location suffix designator	Connector size (no. of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	F Min. Cable opening sq. inch
1 - top opening 2 - side opening	75 Socket only	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (65.89)	2.282 (57.96)	.55 (13.97)
3 - top opening 4 - side opening (see note 7)	75 Socket only	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (65.89)	2.282 (57.96)	.55 (13.97)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.
6. Edges of cable entry hole to be rounded to prevent insulation chafing.
7. Shield and shield clamp location PIN designators,” 3 and 4”, are for use with one-piece jackscrews.
8. Clamp hinge may attach to side of shield.

FIGURE 4. Shield - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



Inches	mm
.001	0.03
.003	0.08
.004	0.10
.120	3.05
.196	4.98
.62	15.75

For use when no shell is specified.

FIGURE 4. Shield - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.

For use when no shell is specified.

Shield clamp location suffix designator	Connector size (no. of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	F Min. Cable opening sq. inch	G ±.047 (1.19 mm)
5 - top opening 6 - side opening	7	1.18 (29.97) 1.35 (34.29)	1.218 (30.94)	.437 (11.10)	.906 (23.01)	.07 (1.78)	.265 (6.73)
5 - top opening 6 - side opening	9	1.18 (29.97) 1.35 (34.29)	1.312 (33.32)	.437 (11.10)	1.000 (25.40)	.16 (4.06)	.265 (6.73)
5 - top opening 6 - side opening	14	1.18 (29.97) 1.35 (34.29)	1.250 (31.75)	.500 (12.70)	.937 (23.80)	.19 (4.83)	.265 (6.73)
5 - top opening 6 - side opening	20	1.18 (29.97) 1.35 (34.29)	1.562 (39.67)	.500 (12.70)	1.250 (31.75)	.22 (5.59)	.265 (6.73)
5 - top opening 6 - side opening	26	1.18 (29.97) 1.35 (34.29)	1.625 (41.28)	.634 (16.10)	1.312 (33.32)	.26 (6.60)	.265 (6.73)

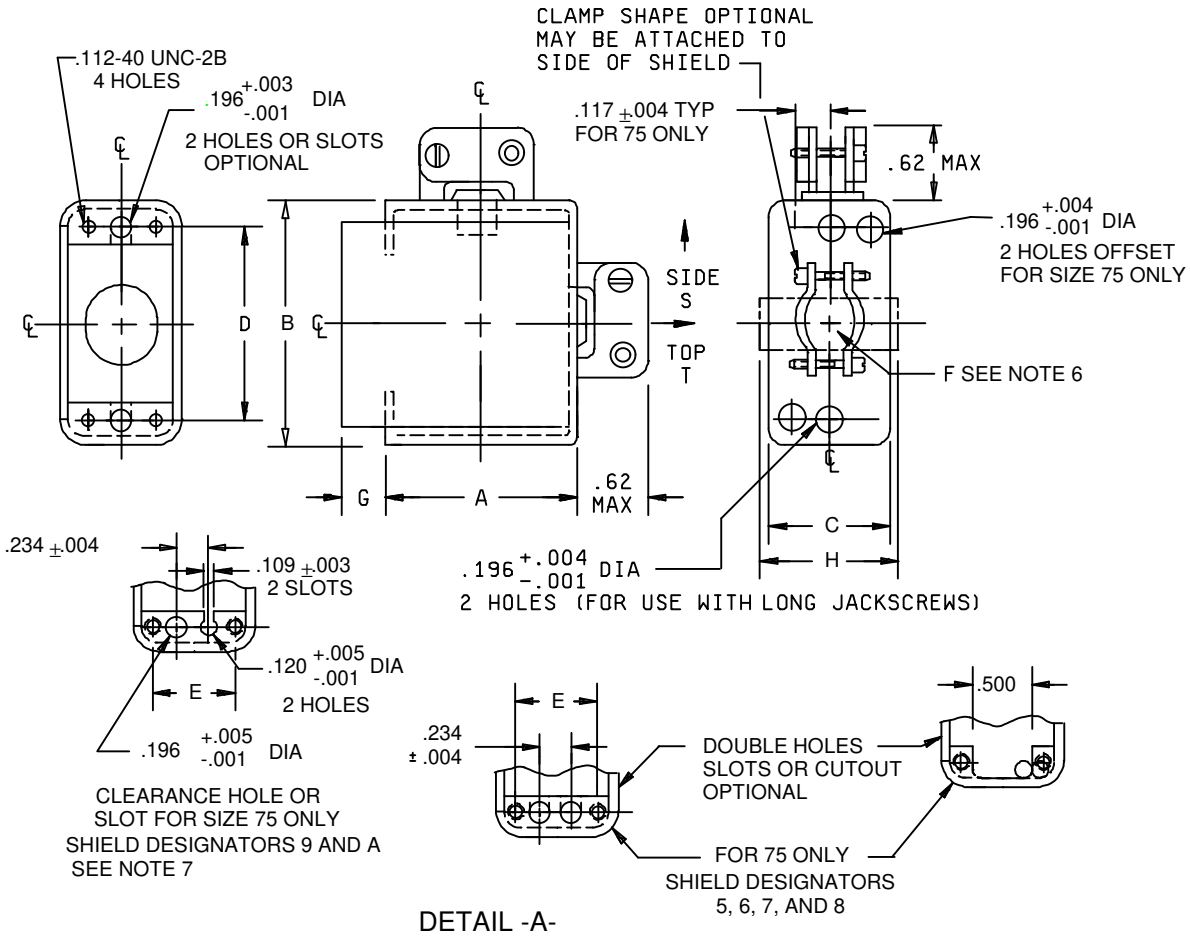
The following PINs are the same as those listed above except the two .120 inch (3.05 mm) through holes for mounting the insert are threaded and the two .196 inch (4.98 mm) holes are omitted. **THESE PARTS ARE NOT RECOMMENDED FOR NEW DESIGN**

Shield clamp location suffix designator	Connector size (no. of contacts)	Thread size
7 - top opening 8 - side opening	7	4-40 UNC-2B
7 - top opening 8 - side opening	9	4-40 UNC-2B
7 - top opening 8 - side opening	14	4-40 UNC-2B
7 - top opening 8 - side opening	20	4-40 UNC-2B
7 - top opening 8 - side opening	26	4-40 UNC-2B

For use when no shell is specified.

FIGURE 4. Shield - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
Check the source to verify that this is the current version before use.



DETAIL -A-

Inches	mm
.001	0.03
.003	0.08
.004	0.10
.005	0.13
.109	2.77
.112	2.84
.117	2.97
.120	3.05
.196	4.98
.234	5.94
.500	12.70
.62	15.70

For use when no shell is specified.

FIGURE 4. Shield - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
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Shield clamp location suffix designator	Connector size (no. of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	E ±.005	F Min. Cable opening sq. inch	G ±.047 (1.19 mm)
5 - top opening 6 - side opening	34	1.22 (30.99) 1.35 (34.29)	2.000 (50.80)	.828 (21.03)	1.168 (29.67)	.468 (11.89)	.34 (8.64)	.265 (6.73)
5 - top opening 6 - side opening	42	1.22 (30.99) 1.35 (34.29)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.468 (11.89)	.34 (8.64)	.265 (6.73)
5 - top opening 6 - side opening	50	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	.828 (21.03)	2.282 (57.96)	.468 (11.89)	.34 (8.64)	.093 (2.36)
5 - top opening 6 - side opening	75	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (30.18)	2.282 (57.96)	.766 (19.46)	.55 (13.97)	.093 (2.36)

The following PIN's are the same as those listed above except the two .196 inch (4.98 mm) thru holes for mounting the insert are threaded and the two .196 inch (4.98 mm) holes are omitted. **THESE PARTS ARE NOT RECOMMENDED FOR NEW DESIGN.**

Shield clamp location suffix designator	Connector size (no. of contacts)	Thread size
7 - top opening 8 - side opening	34	.112-40 UNC-2B
7 - top opening 8 - side opening	42	.112-40 UNC-2B
7 - top opening 8 - side opening	50	.112-40 UNC-2B
7 - top opening 8 - side opening	75	.112-40 UNC-2B

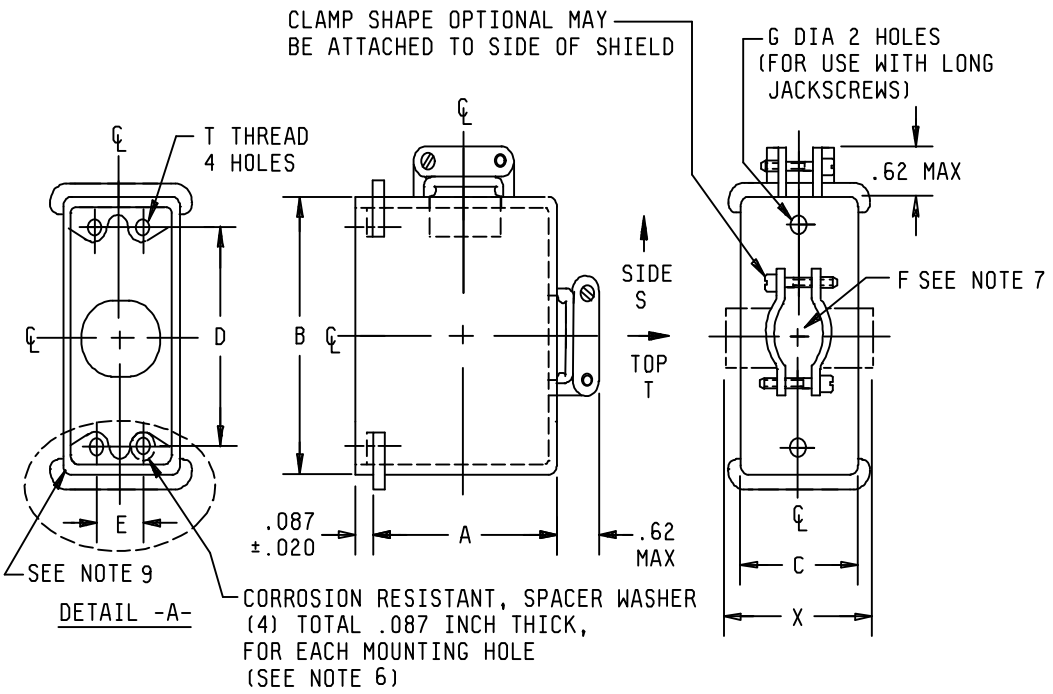
For use with one-piece jackscrews and not used with shells.

Shield clamp location suffix designator (see note 7)	Connector size (no. of contacts)	A min max	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	E ±.005	F Min. Cable opening sq. inch	G ±.047 (1.19 mm)
9 - top opening A - side opening	34	1.22 (30.99) 1.35 (34.29)	2.000 (50.80)	.828 (21.03)	1.168 (29.67)	.468 (11.89)	.34 (8.64)	.265 (6.73)
9 - top opening A - side opening	42	1.22 (30.99) 1.35 (34.29)	2.312 (58.72)	.828 (21.03)	2.000 (50.80)	.468 (11.89)	.34 (8.64)	.265 (6.73)
9 - top opening A - side opening	50	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	.828 (21.03)	2.282 (57.96)	.468 (11.89)	.34 (8.64)	.093 (2.36)
9 - top opening A - side opening	75	1.22 (30.99) 1.35 (34.29)	2.594 (65.89)	1.188 (30.18)	2.282 (57.96)	.766 (19.46)	.55 (13.97)	.093 (2.36)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.16 inch (0.41 mm).
4. Material aluminum alloy in accordance with QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.
6. Edges of cable entry hole to be rounded to prevent insulation chafing.
7. For use with one-piece jackscrews and are not used with shells.

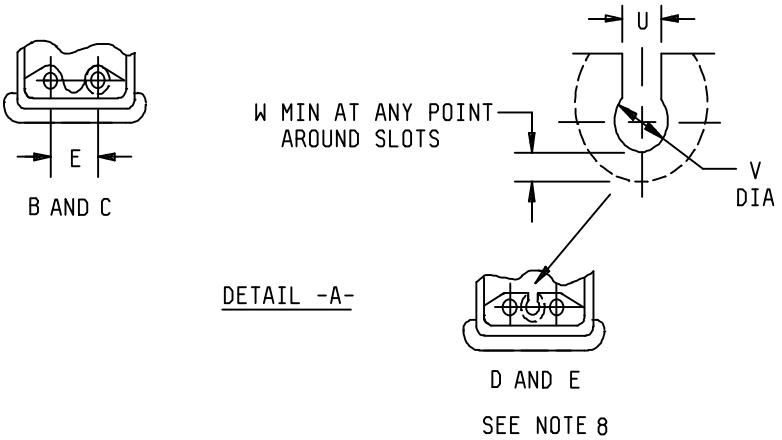
FIGURE 4. Shield - Continued.

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Shield and shield clamp location (see note 8)	U ±.003 (0.08 mm)	V +.005 (0.13 mm) -.001 (0.03 mm)	W Min	Connector size (no. of contacts)
D - top opening E - side opening	.109 (2.77)	.120 (3.05)	.062 (1.57)	66
D - top opening E - side opening	.133 (3.38)	.150 (3.81)	.110 (2.79)	104

Inches	mm
.001	0.03
.003	0.08
.004	0.10
.020	0.51
.087	2.21
.120	3.05
.196	4.98
.62	15.75



Use with or without shells.

FIGURE 4. Shield - Continued.

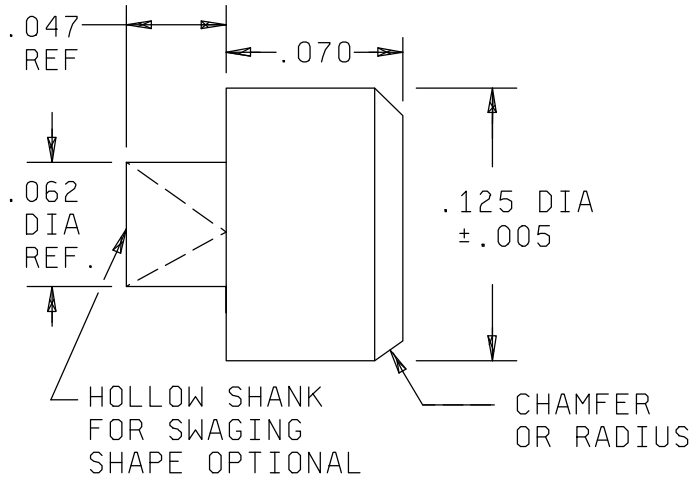
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:10Z
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Shield clamp location suffix designator	Connector size (no. of contacts)	A ±.03 (0.76 mm)	B ± .016 (0.41 mm)	C ± .016 (0.41 mm)	D ±.005 (0.13 mm)	E ±.005 (0.13 mm)	F Min. Cable opening sq. inch	G +.003 (0.08 mm) -.001 (0.03 mm)	T thread	X max
B - top opening C - side opening	66	1.17 (29.72)	2.366 (60.10)	1.211 (30.76)	1.969 (50.01)	.500 (12.70)	.55 (13.97)	.196 (4.98)	.112-40 UNC-2B	1.37 (34.80)
B - top opening C - side opening	104	2.23 (56.64)	2.835 (72.01)	1.625 (41.28)	2.375 (60.33)	.875 (22.23)	1.10 (27.94)	.229 (5.82)	.138-32 UNC-2B	1.88 (47.75)
D - top opening (see note 8) E - side opening	66	1.17 (29.72)	2.366 (60.10)	1.211 (30.76)	1.969 (50.01)	.500 (12.70)	.55 (13.97)	.196 (4.98)	.112-40 UNC-2B	1.37 (34.80)
D - top opening (see note 8) E - side opening	104	2.23 (56.64)	2.835 (72.01)	1.625 (41.28)	2.375 (60.33)	.875 (22.23)	1.10 (27.94)	.229 (5.82)	.138-32 UNC-2B	1.88 (47.75)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.016 inch (0.41 mm).
4. Material aluminum alloy in accordance with QQ-A-250/8, temper H32 .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.
6. Spacer washers, (4), used only with shell, discard in other uses. Spacer washers are corrosion resistant steel.
7. Edges of cable entry hole to be rounded to prevent insulation chafing.
8. For use with one-piece jackscrews.
9. No cracks, chips, or sharp edges on all 4 edges of flange.

FIGURE 4. Shield - Continued.

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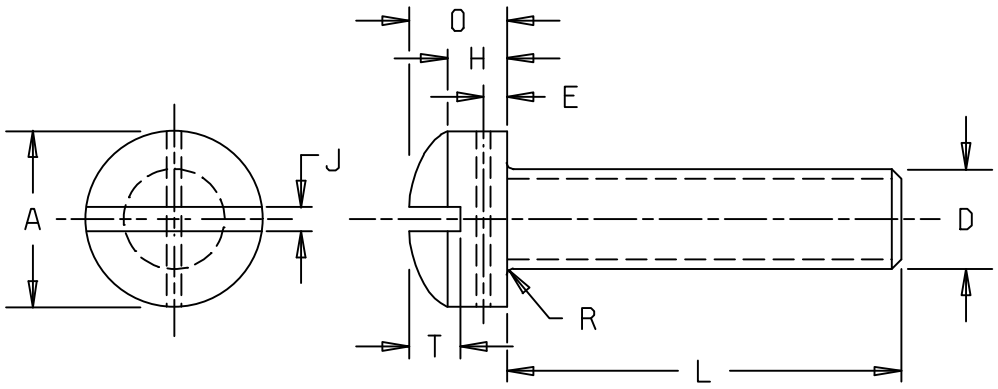
Inches	mm
.005	0.13
.047	1.19
.062	1.57
.070	1.78
.125	3.18

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.016 inch (0.41 mm).
3. Metric equivalents are given for general information only.
4. Material: Corrosion resistant steel in accordance with QQ-S-763, classes 302, 303, 303Se, 304, 316, or ASTM A484, or ASTM A532.
5. See M28748/1 to order polarizing pin separately.

Pin polarizing, straight headed.

FIGURE 5. Pin configuration.

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Screws furnished with shields when required crossed drilled hole is optional.
To order screws individually use MS35275-215 (#4) or -229 (#6).

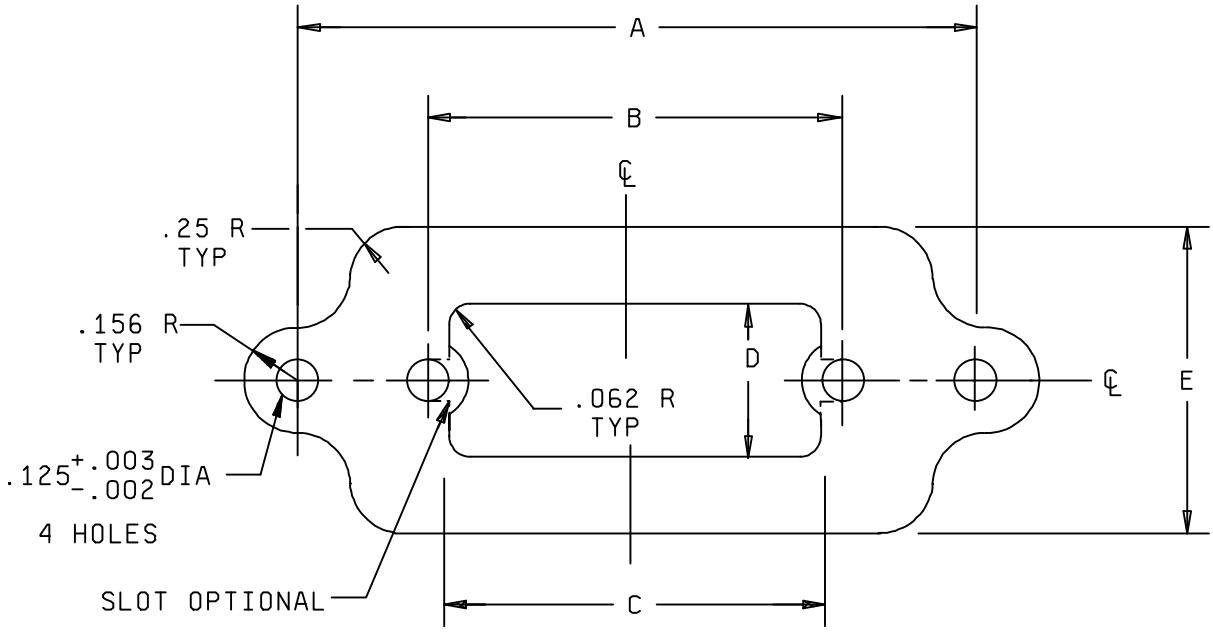
Shield designator	Nominal size	Threads per inch, UNC-2A (Pitch)	D Body diameter min	D Body diameter max	A Head diameter min	A Head diameter max	H Head side height min	H Head side height max	O Total head height min	O Total head height max
1, 2, 3, 4, (sizes 34, 42, 50, 75); 5, 6, 7, 8, (sizes 34, 42, 50, 75); 9, A, B, C, D, E, (size 66)	.112 (#4) (2.845)	40 (.64)	.0925 (2.350)	.1120 (2.845)	.166 (4.216)	.183 (4.648)	.072 (1.829)	.079 (2.007)	.094 (2.388)	.107 (2.718)
B, C, D, E size 104 only	.138 (#6) (3.505)	32 (.79)	.1141 (2.898)	.1380 (3.505)	.208 (5.283)	.226 (5.740)	.089 (2.261)	.096 (2.438)	.118 (2.997)	.132 (3.353)

Shield designator	J Slot width min	J Slot width max	T Slot depth min	T Slot depth max	E drill location nom	Drilled hole size nom	R radius max	Tensile strength LB-min (KN-min)
1, 2, 3, 4, (sizes 34, 42, 50, 75); 5, 6, 7, 8, (sizes 34, 42, 50, 75); 9, A, B, C, D, E, (size 66)	.031 (.787)	.039 (.991)	.030 (.762)	.038 (.965)	.035 (.889)	.037 (.940)	.017 (.432)	.480 (2.151)
B, C, D, E size 104 only	.039 (.991)	.048 (1.219)	.035 (.889)	.045 (1.143)	.043 (1.092)	.046 (1.168)	.021 (.533)	.730 (3.272)

NOTES:
1. Material corrosion resisting, in accordance with Federal standard number 66, AISI 302, 303, 304, 305, 316, 384 (developed for cold heading) or type UNS S30430 in accordance with the chemical requirements of ASTM-A-493.
2. Treatment: Passivation shall be in accordance with QQ-P-35, or an industry approved equivalent.
3. Magnetic permeability: These screws shall have a magnetic permeability or 2.0 maximum (air = 1.0) for a field strength of H = 200 oersteds, using a magnetic indicator in accordance with ASTM A342.
4. Threads: Threads shall be in accordance with FED-STD-H28.
5. Threads: Screws shall have complete threads extending to within two threads of the head, or closer if practicable.
6. Drilled hole is to be approximately at right angles to the slot and may be permitted to break through the bottom of the slot. All sharp edges shall be removed.

FIGURE 6. Slotted screw.

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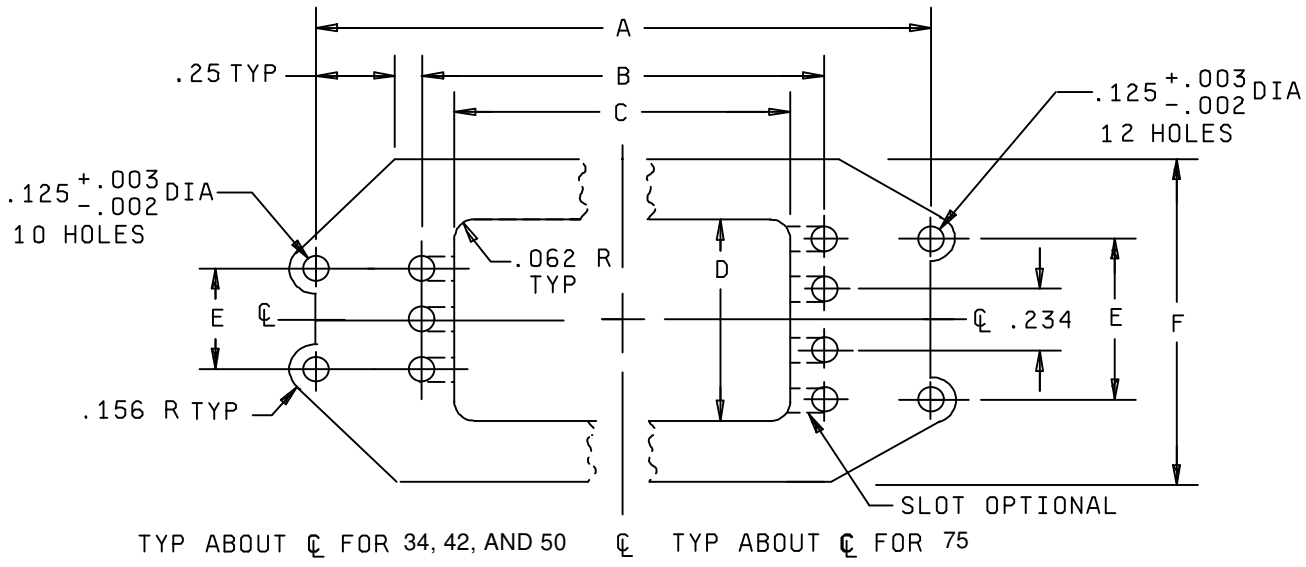


Inches	mm
.002	0.05
.003	0.08
.062	1.57
.125	3.18
.156	3.96
.25	6.35

Connector size (No. of contacts)	A ±.005 (0.13 mm)	B ±.005 (0.13 mm)	C	D	E
9	2.016 (51.21)	1.000 (2.54)	.875 (22.23)	.406 (10.31)	.953 (24.21)
14	1.953 (49.61)	.937 (23.80)	.812 (20.62)	.469 (11.91)	1.016 (25.81)
20	2.266 (57.56)	1.250 (31.75)	1.125 (28.58)	.469 (11.91)	1.016 (25.81)

FIGURE 7. Retaining plate.

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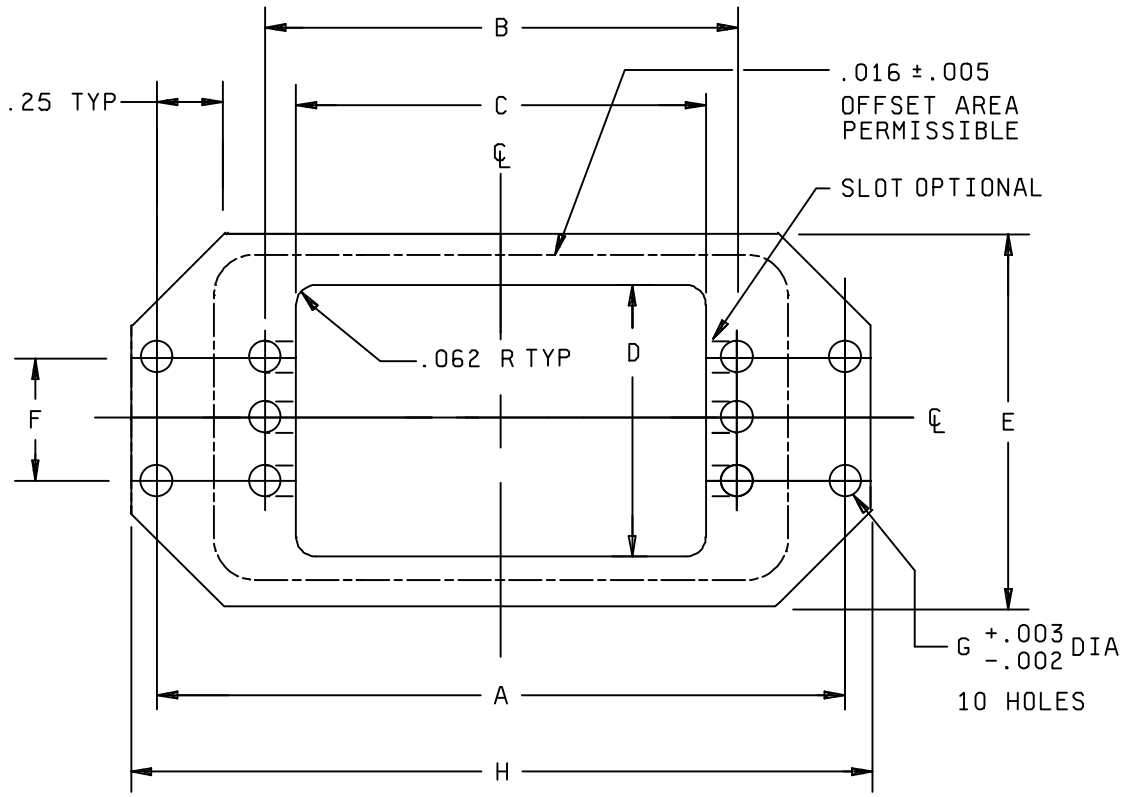


Inches	mm
.002	0.05
.003	0.08
.062	1.57
.125	3.18
.156	3.96
.234	5.94
.25	6.35

Connector size (No. of contacts)	A ±.005 (0.13 mm)	B ±.005 (0.13 mm)	C	D	E ±.005 (0.13 mm)	F
34	2.703 (68.66)	1.687 (42.85)	1.438 (36.53)	.750 (19.05)	.468 (11.89)	1.328 (33.73)
42	3.016 (78.89)	2.000 (50.80)	1.750 (44.45)	.750 (19.05)	.468 (11.89)	1.328 (33.73)
50	3.422 (86.92)	2.281 (57.94)	2.031 (51.60)	.750 (19.05)	.468 (11.89)	1.453 (36.91)
75	3.422 (86.92)	2.281 (57.94)	2.031 (51.60)	1.109 (28.17)	.766 (19.46)	1.797 (45.64)

FIGURE 7. Retaining plate - Continued.

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Inches	mm
.002	0.05
.003	0.08
.062	1.57
.125	3.18
.156	3.96
.234	5.94
.25	6.35

Connector size (No. of contacts	A ±.005 (0.13 mm)	B ±.005 (0.13 mm)	C	D	E	F ±.005 (0.13 mm)	G	H
66	2.984 (75.79)	1.969 (50.01)	1.718 (43.64)	1.125 (28.58)	1.688 (42.88)	.500 (17.70)	.125 (3.18)	3.296 (83.72)
104	3.500 (83.90)	2.375 (60.33)	2.125 (53.98)	1.486 (37.74)	2.000 (50.80)	.875 (22.23	.150 (3.81)	3.875 (98.43)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerances are ±.16 inch (0.41 mm).
4. Material aluminum alloy per specification QQ-A-250/8, temper H32, .040 inch (1.02 mm) thick.
5. Finish anodize in accordance with MIL-A-8625, type I or type II.

FIGURE 7. Retaining plate - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, 3, 4, 5, 6, and 7.

Mating connector: See MIL-DTL-28748/1 and MIL-DTL-28748/3.

PIN: As shown in the following example: 1/

(See (PIN) code)

M28748/2	B	A	1	H	L	1 A
Basic number of specification sheet	Hardware designator (see figure 1 for configuration styles)	Insert designator (see figure 2)	Shield and shield clamp location or retaining plate (see figure 4 or 7)	Shell and polarization position (see figure 3)	Jackscrews or guidepins (see figure 1)	Contacts

PIN code:

Hardware Designators 2/	Insert designators	Number of contact positions	Shield and shield clamp Locations	Shell and Polarization designators	Polarization positions 3/
				0 - None included	
A - Long jackscrew hardware only.	A	9	1 - top opening 5/	A	A
B - All other hardware.	B	14	2-side opening 5/	B	B
- - (insert only)	C	20	3 - top opening 6/	C Plug	C
	D	34	4-side opening 6/	D shells --	D
	E	42	5 - top opening 7/	E	E
	F	50	6-side opening 7/	F	F
	G	66	7- top opening 8/	G	G
	H	75	8- side opening 8/	H - Unpolarized	
	J	104	9 - top opening 9/	J	A
	0	No insert included	A-side opening 9/	K	B
			B - top opening 10/	L Receptacle	C
	Insert designators through hole 4/		C - side opening 10/	M shells ----------	D
			D- top opening 11/	N	E
	K	34	E - side opening 11/	P	F
	L	42		Q	G
	M	50	0-none included	R - Unpolarized	
	N	66		S	A
	P	75	Retaining plate	T	B
	Q	104	P - Plate	U Plug	C
				V shells --	D
				W	E
				X	F
	Insert designators for piece part orders 12/	Connector sizes	Through hole insert designators for piece part orders 13/	Y	G
				Z - Unpolarized	
				1	A
				2	B
				3 Receptacle	C
	R	9	|	4 shells ----------	D
	S	14	|	5	E
	T	20	|	6	F
	U	34	1	7	G
	V	42	2	8 - Unpolarized	
	W	50	3		
	X	66	4		
	Y	75	5		
	Z	104	6		

See footnotes on next page.

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Jackscrews
or
guidepins

F - Fixed jackscrews, configuration C, figure 1 (not used with shield).
G - Guidepins, configuration B, figure 1.
L - Long jackscrews, one-piece construction, configuration E, figure 1.
M - Long jackscrews, multi-piece construction, configuration F, figure 1.
S - Short jackscrews slotted head, configuration D, figure 1 (not used with shield).
0 - None included.

Jack and guide location: The jackscrew male (turning or fixed) or guide pin shall be located by the A or 1 pin contact.

Contacts: In accordance with MIL-DTL-28748.

1A - 100 percent size 16
1L - none included 13/

1/ The number zero (0) is used to indicate which parts are not included.
2/ Used for information to indicate size and style of insert or hardware to be used.
3/ Polarization positions are for location of plug and receptacle shells pins and slots (see figure 3).
4/ Manufacturers, mold bodies, and hardware will not necessarily interchange with each other.
5/ Used when shell is specified and with “L” hardware or used with or without shell for “M” hardware.
6/ Use when no shell is specified and with “L” hardware.
7/ Use when no shell is specified and with “M” hardware.
8/ Use when no shell is specified and with “G” hardware, not for new design.
9/ Used when no shell is specified and with “L” hardware.
10/ Use with or without shell for “M” hardware or used when shell is specified and with “L” hardware.
11/ Used when no shell is specified and with “L” hardware.
12/ These designators are informational and used for insert reference (connector size) to order an appropriate shield or shell as a piece part. Shields and shells may be ordered together or separately.
13/ The “1L” ordering code shall be used in conjunction with insert designator zero “0” or with the insert designators for piece part orders for which no insert is included.

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TABLE I. Superseded specification and MS sheets.

Specification or MS sheet	Dated
MIL-C-28748/2D	30 November 1981
MS18173	20 May 1970
MS18192	1 June 1983
MS18193	26 April 1990
MS18198	23 June 1972
MS18199	23 June 1972
MS18200	20 May 1970
MS18203	26 April 1990
MS18204	26 April 1990
MS18205	23 June 1972
MS18239	23 June 1972
MS18241	23 June 1972
MS18243	23 June 1972
MS18245	23 June 1972
MS18247	23 June 1972
MS18249	23 June 1972
MS18251	23 June 1972
MS18253	23 June 1972
MS18255	23 June 1972
MS24132	1 June 1976
MS24133	3 January 1976
MS35275	14 June 1976
MS35275	14 June 1976

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4132-02)

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 17, 99

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
3 The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-28748/2D	**2. DOCUMENT DATE** *(YYYYMMDD)* 1999 May 11

3. DOCUMENT TITLE
CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, SOLDER TYPE SOCKET CONTACTS, SIZE 16

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include zip code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME **Defense Logistics Agency** **Defense Supply Center, Columbus**	b. TELEPHONE *(Include Area Code)* (1) Commercial **614-692-0573** (2) DSN **850-0573**
c. ADDRESS *(Include Zip Code)* **DSCC-VAI** **P.O. Box 3990** **Columbus, Ohio 43216-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Standardization Program Office (DLSC-LM)** **8725 John J. Kingman Road,Suite 2533** **Fort Belvoir, Virginia 22060-6621** **Telephone (703) 767-6888 DSN 427-6888**

DD Form 1426, FEB 1999 (EG) PREVIOUS EDITION IS OBSOLETE. WHS/DIOR, Feb 99

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